

A transdiagnostic dimensional map as a self-calibrating measurement instrument: projection, information, and value of information under structured missingness

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Abstract

Dimensional maps of psychopathology are only as trustworthy as the measurement that produces them, yet they are typically read as fixed point-coordinates that hide how sharply each patient is located, how much each instrument contributed, and how cheaply the map could be measured better. We present the transdiagnostic dimensional map as a *measurement instrument* rather than a finding. A missingness-native Bayesian bifactor / exploratory structural-equation model — fit to each patient’s observed cells only (no imputation), under mixed likelihoods and a Gaussian-copula continuous block — returns for every one of $N = 9,013$ patients (bipolar disorder, schizophrenia, major depression; 21 sites) a posterior coordinate *and* its uncertainty ellipsoid, and its closed-form structure makes four instrument properties computable on that one object. **Projection:** an expected-a-posteriori score map, reduced to the eight-dimensional factor space by a Woodbury identity, returns the prior — never a false-precise point — on unmeasured axes. **Information:** because indicators are conditionally independent given the factors, posterior precision accumulates additively, and the quantity that governs localization is the Fisher information each indicator contributes at its own likelihood family, *not* its loading — high-loading, low-prevalence flags contribute almost nothing. **Adaptive sampling:** a loading-dependent minimal-indicator rule recovers most of each axis’s reliable signal in three to six items and exposes two axes that are intrinsically capped by the instrument bank (mania/activation 0.41, substance 0.43) rather than by missingness; a fully patient-adaptive selection order coincides with the fixed order, showing the population-mean reliability curve is a *tight* upper bound. **Value of information:** the entropy reduction from a not-yet-measured indicator is closed-form (matrix-determinant lemma), so a greedy rule assembles a shared 27-item battery at mean reliability 0.70 and localizes where each cohort is under-measured. We show by simulation that the posterior coordinate is well-calibrated (empirical coverage 0.95 matches nominal, flat across observed-item counts and including the bank-limited axes) and that its calibration degrades gracefully rather than collapsing under four violations of the model’s assumptions; that the shrinkage-corrected posterior-mean coordinate improves downstream prognosis over a naive sum-score, most under sparse measurement (while the posterior’s uncertainty does not, on its own, triage who is predictable for a coarse remission endpoint); and that ranking items by information rather than loading changes which battery you would build (+0.11 reliability at 27 items). A five-archetype convex-blend simplex summarizes the continuum — which has no natural clusters — as graded membership in five named clinical poles, locating a patient $\approx 9.7\times$ more tightly than the DSM-5 label; we show its modest reconstruction fidelity (59% of coordinate variance) is the wrong yardstick for it, since the near-isotropic

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variance lets even a random five-dimensional subspace retain more, and the simplex earns its place by interpretability, not variance retained. The contribution is the instrument: a map that reports its own uncertainty, weighs instruments by the information they carry, and prescribes how to measure the next cohort more cheaply. The clinical and biological findings this map supports are reported in a companion paper; here the map’s biology serves only as a worked example of the instrument.

Keywords: measurement model; item response theory; Fisher information; computerized adaptive testing; value of information; missing data; bifactor model; archetypal analysis; posterior calibration; transdiagnostic psychiatry.

1 Introduction

Dimensional models of psychopathology are displacing categorical diagnosis as the organizing frame for clinical research. Hierarchical and general-factor accounts [1–4], normative and biotype modelling [5–7], and the transdiagnostic ambition of the Research Domain Criteria [8] all reduce a patient to a position in a low-dimensional latent space. That position is then read, plotted, correlated with outcomes, and increasingly proposed as a target for stratified care.

A dimensional coordinate, however, is an *estimate*, and an estimate without its uncertainty is a claim without an error bar. In practice the measurement step that produces the coordinate is treated as a solved preliminary: incomplete assessments are imputed or dropped, mixed item types are coerced to a common scale, and each patient is reported as a point. This hides exactly the quantities a user of the map needs. How sharply is a given patient actually located — is a coordinate pinned down by a dozen observations or floated in from the prior on a single item? Which of the administered instruments earned that precision, and which contributed almost nothing? If the assessment is incomplete, what is the single most valuable thing left to measure? And when a coordinate is later fed into a prognostic model or a stratification rule, does its uncertainty travel with it, or is a floated-in prior mean treated as though it were pinned down by data? Point-estimate, imputation-based scoring cannot answer any of these; the questions are not about the map’s content but about the *quality of its measurement*.

The components needed to answer them already exist, each well characterised in isolation. Item response theory gives the likelihood of a mixed-type response given a latent position and, through Fisher information, a currency for how much each item constrains that position [9–12]. Expected-a-posteriori scoring and the Woodbury identity make posterior coordinates and their covariance computable at scale [13, 14]. Computerized adaptive testing turns information into a sequential item-selection rule [15]; value-of-information reasoning turns it into a design criterion [16]; a principled treatment of missingness [17, 18] makes it legitimate to score a patient from observed cells alone; and archetypal analysis [19] gives an interpretable low-dimensional summary of a continuous coordinate cloud. What has not been done is to assemble these into a single object and validate them *together* on a real, pervasively incomplete, mixed-type transdiagnostic assessment — and then to read the resulting map not as a set of coordinates but as an instrument that can be interrogated about its own measurement.

Here we do exactly that. Using one missingness-native Bayesian bifactor / exploratory structural-equation model, fit to the harmonised baseline assessment of $N = 9,013$ patients with bipolar disorder, schizophrenia or major depression from the three FACE cohorts, we treat the eight-dimensional dimensional map as a measurement instrument and ask four operational questions of it. *Where is a patient, and how sharply* (projection)? *Which items earned that precision* (information)? *How few items suffice, and where is the instrument bank itself the limit* (adaptive sampling)? *What is the cheapest next measurement, and which cohorts are under-measured* (value of information)? We then stress-test the instrument where a measurement reviewer will push hardest: we show by simulation that its credible intervals are calibrated, and we quantify how faithfully its five-archetype convex-blend summary reconstructs the full

coordinate.

The claim of this paper is therefore methodological, not clinical: *a transdiagnostic dimensional map is not a fixed set of point-coordinates but a self-calibrating measurement instrument* — one that returns, for every patient, a posterior coordinate and its uncertainty, quantifies how much information each indicator contributed (Fisher information, not loading), prescribes the cheapest next measurement that shrinks that uncertainty, and propagates that uncertainty into every downstream use. The clinical and biological findings that this map supports — including the immunometabolic axis that recurs as its own archetype corner — are the subject of a companion paper and are deliberately out of scope here; where a specific patient or a specific axis appears below, it is an illustration of an instrument property, never a substantive claim about psychopathology.

2 Methods

We describe the instrument in the order a reader uses it: the measurement model and the no-imputation data layer it rests on (§2.1); the loading read as a measurement slope (§2.2); projection — EAP scoring and the Woodbury reduction — which turns a patient’s observed cells into a posterior coordinate and covariance (§2.3); the information accounting that weighs each indicator (§2.4); adaptive sampling and value of information (§2.5); archetypes as a low-dimensional summary (§2.6); and the calibration protocol that tests whether the instrument’s uncertainty is honest (§2.7). The model, its estimation, and its confirmatory checks are those of the companion paper and are recapitulated here only as far as the instrument properties require. The Supplementary Material gives the full derivations — the measurement model and identifiability (Supplement S1), the Woodbury projection and EAP scoring (S2), the loading-as-slope (S3), the per-family Fisher information and reliability accounting (S4), the value-of-information and adaptive-sampling results (S5), the archetype reconstruction (S6), and the calibration protocol (S7) — with every proposition proved.

2.1 The measurement model and the observed-cell invariant

Each patient i carries a latent coordinate $f_i \in \mathbb{R}^{K+1}$ — a general functional-burden factor G and $K = 7$ specific dimensions (cognition, immunometabolic load, sleep, mania/activation, suicidality, developmental risk, substance use) — with prior $f_i \sim \mathcal{N}(\mathbf{0}, \Phi)$. Each indicator j is a linear reflection of those factors through a single linear predictor,

$$\eta_{ij} = \alpha_j + \lambda_{jG} G_i + \sum_{k=1}^K \lambda_{jk} D_{ik} + \beta_j^\top c_i, \quad (1)$$

with item intercept α_j , general loading λ_{jG} , specific loadings λ_{jk} (row j of $\Lambda = [\lambda_G \mid \Lambda_D]$), and item-level covariates c_i . Equation (1) is the bifactor exploratory structural-equation model [20, 21]. The predictor feeds a per-type observation density — Gaussian or log-Gaussian for continuous and laboratory measures, ordered-logit for ordinal, Bernoulli for binary, negative-binomial for counts — and continuous indicators additionally enter through a rank-based Gaussian-copula marginal $z_{ij} = \Phi_{\mathcal{N}}^{-1}(\hat{F}_j(X_{ij}))$, so that placing heterogeneous variables on a shared scale never manufactures covariance. The fitted map carries 109 indicators across the eight axes — 88 in the Gaussian-copula continuous block and 21 in an explicit non-Gaussian block of Bernoulli, ordinal and count items — so that all four likelihood families are represented and the information accounting of §2.4 must reckon with each.

Given the latent scores the indicators are conditionally independent (local independence): once the factors are fixed, only item-specific noise remains, with diagonal residual covariance Ψ . This single property is what makes the map an instrument, because it makes the factors the

only source of shared covariance, so the model-implied covariance is

$$\Sigma = \Lambda \Phi \Lambda^\top + \Psi, \quad (2)$$

and it is also why imputation is forbidden: a filled-in cell would inject covariance that (2) would misread as a factor. Accordingly the model is fit to each patient’s *observed cells only*. With F_j the density of indicator j and O_i the set of cells patient i actually has, the observed-data log-likelihood is

$$\log p(X_{\text{obs}} | \Theta) = \sum_{i=1}^N \sum_{j \in O_i} \log F_j(X_{ij} | \eta_{ij}, \theta_j). \quad (3)$$

A missing cell contributes no term and no completed $N \times J$ matrix is ever formed. For the Gaussian block this is exact full-information maximum likelihood [17, 18]: marginals of a normal are read off by deleting coordinates, so marginalizing the missing cells is the marginal property itself, not an imputation step. Equation (3) is the no-imputation invariant on which every instrument property below rests.

The primary bifactor fit holds G orthogonal to the specific axes; a correlated- G arm instead estimates the cross-correlations

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^\top \\ \phi_{Gd} & \Phi_D \end{pmatrix}, \quad \phi_{Gd} = \text{Corr}(G, D) \in [-1, 1]^K, \quad (4)$$

so that whether a domain is entangled with severity becomes a measured quantity rather than an imposed constraint. The model is fit globally at full sample ($N = 9,013$) under per-patient cohort weights $w_i = N/(3n_{c(i)})$; sampling used the No-U-Turn sampler [22] in PyMC [23] with the NumPyro/JAX backend [24]. Estimation, convergence gating, measurement-invariance and robustness are as in the companion paper.

2.2 The loading as a measurement slope

Equation (1) separates two intrinsically different objects. The factor f_i is a per-patient latent *position*; a loading is a fixed *slope* — the rate at which a measurement changes as that position moves, shared by every patient. On the latent linear-predictor scale this slope is exact and patient-independent, $\partial \eta_{ij} / \partial f_{ik} = \lambda_{jk}$, so the loading matrix is the Jacobian of the measurement with respect to the latent. On the response scale, for the single-index likelihoods with conditional mean $\mathbb{E}[X_{ij} | f_i] = h_j(\eta_{ij})$, the chain rule gives the marginal effect of a factor on an indicator and its population average,

$$\frac{\partial \mathbb{E}[X_{ij} | f_i]}{\partial f_{ik}} = h'_j(\eta_{ij}) \lambda_{jk}, \quad s_{jk} \equiv \mathbb{E} \left[\frac{\partial \mathbb{E}[X_{ij} | f_i]}{\partial f_{ik}} \right] = \kappa_j \lambda_{jk}, \quad \kappa_j = \mathbb{E} [h'_j(\eta_{ij})] \geq 0. \quad (5)$$

The loading is thus how strongly an item *reflects* a factor. It is not, as §2.4 shows, how much that item *localizes* a patient — and the distinction between the two is the paper’s central measurement point.

2.3 Projection: EAP scoring and the Woodbury reduction

The evaluation of (2) naively costs $\mathcal{O}(|C_i|^3)$ per patient. The Woodbury identity reduces it to the $(K+1)$ -dimensional factor space,

$$\Sigma_{C_i}^{-1} = \Psi_{C_i}^{-1} - \Psi_{C_i}^{-1} \Lambda_{C_i} (\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i})^{-1} \Lambda_{C_i}^\top \Psi_{C_i}^{-1}, \quad (6)$$

whose inner $(K+1) \times (K+1)$ solve is independent of the number of observed cells $|C_i|$ and is shared by all patients with the same missingness pattern (computed once per pattern, with the

matrix-determinant lemma for the log-determinant). This is what makes full-sample projection tractable [14]. Each patient’s coordinate is then the expected-a-posteriori (EAP) factor score from their observed cells; for an all-continuous pattern,

$$\hat{f}_i = \mathbb{E}[f_i | X_{i,C_i}] = \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} (X_{i,C_i} - \mu_{i,C_i}), \quad S_i = \Phi - \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} \Lambda_{C_i} \Phi, \quad (7)$$

obtained by Markov-chain Monte Carlo when non-Gaussian cells are present [13]. The posterior mean $x_i \equiv \hat{f}_i \in \mathbb{R}^8$ and the posterior covariance S_i are the two objects every downstream analysis consumes. The structure of (7) makes the instrument’s honesty mechanical: on an axis with no observed indicator the corresponding rows of Λ_{C_i} are empty, the update term vanishes, and the posterior returns the prior $\mathcal{N}(0, \Phi)$ — a wide interval, never a false-precise point. A patient with one observed indicator on an axis is not treated as equally characterised as one with six.

2.4 Information accounting: what localizes a patient

The posterior covariance S_i in (7) inverts a precision matrix. Under local independence that precision is additive over observed items: from (6),

$$S_i^{-1} = \Phi^{-1} + \sum_{j \in O_i} \mathcal{I}_j(f_i), \quad (8)$$

where $\mathcal{I}_j$ is the Fisher information item j contributes about the factors. Each observed cell adds a non-negative term; nothing an item measures is ever subtracted. For a single-index likelihood the information item j carries about factor k is

$$\mathcal{I}_{jk}(f_i) = \lambda_{jk}^2 \mathcal{J}_j(\eta_{ij}), \quad (9)$$

the squared loading times a family-specific unit information $\mathcal{J}_j$ evaluated at the item’s linear predictor: for a Gaussian item $\mathcal{J}_j = 1/\sigma_j^2$ (independent of f_i); for a Bernoulli item $\mathcal{J}_j = p_{ij}(1 - p_{ij})$ with $p_{ij} = \text{logit}^{-1}(\eta_{ij})$; for a negative-binomial count $\mathcal{J}_j = \mu_{ij} r_j / (r_j + \mu_{ij})$ with $\mu_{ij} = e^{\eta_{ij}}$; and for a graded-response ordinal item the standard sum $\sum_c (\partial_\eta P_{ijc})^2 / P_{ijc}$ over category probabilities [10, 11]. Reliability of an axis score is the familiar function of accumulated precision, $\rho_k = 1 - 1/(1 + \sum_j \mathcal{I}_{jk})$, taking the standardized-factor prior precision as 1.

Equation (9) is why loading is not information. The loading λ_{jk} enters squared, but it is multiplied by a unit information that collapses for a rare binary flag: a Bernoulli item endorsed by 1% of patients contributes $\mathcal{J}_j = p(1 - p) \approx 0.01$ regardless of how strongly it loads, so a high-loading, low-prevalence indicator localizes almost nobody. The quantity that governs how sharply a patient sits on an axis is therefore (9) evaluated per likelihood family, not the loading read off the factor table — the lens §3.2 adopts.

2.5 Adaptive sampling and value of information

Ordering the candidate items for an axis by their information (9) and accumulating (8) gives a reliability-versus-items curve; the number of items to reach a target reliability is a loading-dependent minimal-indicator rule. Evaluated at the population mean $f_i = \mathbf{0}$ this curve is the efficiency of a fixed, non-adaptive most-informative-first battery, and it is an *upper bound* on the efficiency of a computerized adaptive test (CAT) that re-evaluates (9) at each patient’s provisional estimate [15], because the CAT does not know f_i in advance. To ask how loose that bound is, we simulate the CAT directly (§3.3): for each of 500 simulated patients per axis (true $\theta \sim \mathcal{N}(0, 1)$, seed 20240704), starting from the prior we repeatedly select the highest-information not-yet-administered item at the current grid-EAP estimate $\hat{\theta}$, simulate its response from the model at the patient’s true θ , update $\hat{\theta}$ on a 401-point grid, and accumulate the selection-time information. The realized reliability-versus-items curve, averaged over patients, is compared to the fixed-order upper bound on identical single-axis footing.

The same information turns into a design criterion across axes. Because S_i^{-1} is a precision, the entropy of the posterior coordinate is $\frac{1}{2} \log \det(2\pi e S_i)$, and adding a candidate item j^* updates the determinant by the matrix-determinant lemma,

$$\log \det S_i^{-1}|_{+j^*} - \log \det S_i^{-1} = \log(1 + \lambda_{j^*}^\top \mathcal{J}_{j^*} S_i \lambda_{j^*}), \quad (10)$$

a closed-form entropy reduction that *depends on what has already been measured* through S_i [14, 16]. Greedily selecting the item with the largest (10), cycling across the eight axes, assembles a shared cross-axis battery whose value is exact at each step (§3.4); the same accounting, applied per cohort, localizes exactly where an axis is under-measured.

2.6 Archetypes as a low-dimensional summary

The cloud of posterior coordinates $\{x_i\}$ is a graded continuum rather than a set of discrete biotypes (a single-Gaussian null reproduces the apparent clustering; §3.6). We summarize it, without imposing clusters, by archetypal analysis [19], which writes each patient as a convex blend of A extreme phenotypes that are themselves convex combinations of patients,

$$\min_{W,B} \sum_{i=1}^N \left\| x_i - \sum_{a=1}^A w_{ia} z_a \right\|^2 \quad \text{s.t.} \quad z_a = \sum_{j=1}^N b_{ja} x_j, \quad w_{ia}, b_{ja} \geq 0, \quad \sum_a w_{ia} = \sum_j b_{ja} = 1, \quad (11)$$

i.e. $X \approx WZ$ with $Z = BX$ and W, B row-stochastic. The simplex weight $w_i \in \Delta^{A-1}$ is a patient’s soft membership. The number of corners ($A = 5$) is the largest with cross-seed Tucker congruence ≥ 0.8 (a stability cliff at $A = 6$), set by stability rather than parsimony, and w_i is propagated by refitting across posterior draws. Because (11) is a rank- A convex approximation, it discards information; §3.6 quantifies how much by reconstructing $\hat{x}_i = \sum_a w_{ia} z_a$ and reporting the per-axis and pooled coefficient of determination against the full coordinate, with a 5-cluster k -means partition and a 5-component principal-component projection as the interpretable-floor and linear-optimum baselines.

2.7 Calibration protocol

An instrument that reports uncertainty must be tested for whether that uncertainty is honest. Because the truth is unknown for real patients, we test coverage by simulation. We draw 10,000 synthetic patients (seed 20260704) with known latent coordinate $\theta \sim \mathcal{N}(0, \Phi)$ under the frozen model’s own factor prior, generate each item’s response from its fitted family at $\eta = \alpha_j + \lambda_j^\top \theta$, and impose a real missingness pattern by bootstrapping the observed/missing masks of the 9,013 real patients (only the binary observed-or-missing indicator is used, never any clinical value). Each synthetic patient is projected through the same EAP machinery by Fisher-scoring (a Gauss–Newton/Laplace step using the expected information (9)), which reduces exactly to the closed form (7) on all-Gaussian patterns (verified to 3×10^{-15}) and generalises it to the non-Gaussian families. We then measure, per axis and at nominal levels $\{0.50, 0.80, 0.90, 0.95\}$, the empirical fraction of synthetic patients whose true θ_k lies within $\hat{x}_{ik} \pm z [S_i]_{kk}^{1/2}$, the joint coverage of the 8-dimensional credible ellipsoid via the Mahalanobis distance against a χ_8^2 quantile, and the 95% coverage stratified by the number of observed indicators on the axis. A correctness check falls out by construction: on an axis with zero observed items the projection returns the prior, so coverage must equal nominal — prior in, prior out.

2.8 Comparative validation: baselines, downstream utility, and robustness

Four analyses test the instrument against the simpler tools a reader already has, and against violations of its own assumptions. *(i) Battery rule.* We rebuild the shared battery of §2.5 a second way, ranking the same pool of (item, axis) candidates by loading magnitude $|\lambda_{jk}|$

instead of by Fisher information, and compare mean reliability across the eight axes at matched battery sizes; both rules accrue the same additive precision (8), so the comparison isolates the ranking key. *(ii) Reconstruction anchors.* To place the archetype reconstruction R^2 on an interpretable scale we compute, on the same coordinate cloud, the variance retained by the raw eight-dimensional coordinate (1 by definition), a five-component PCA (0.80), a dimension-matched four-component PCA, a five-centroid k -means partition, a single centroid (0), and the mean over 500 Haar-random five-dimensional subspaces, whose expectation is the analytic $k/d = 5/8$; archetype counts $A \in \{4, 5, 6\}$ are refit to check stability. *(iii) Downstream prognosis.* On items simulated from the frozen model (as in the calibration protocol) with a fixed logistic latent-to-remission rule $\text{logit } p_i = w^\top \theta_i$, we compare two estimators as predictors of the held-out outcome: the full EAP coordinate, and a naive per-axis sum-score (the sign-oriented mean of observed items whose primary axis is k , standardized). Both feed an identical cross-validated logistic model; we sweep item completeness from full to 12% and repeat over twelve seeds and a second weight vector. A parallel test on the *real* cohort regresses observed remission on the eight EAP coordinates and asks whether triaging by posterior uncertainty (mean per-axis $[S_i]^{1/2}$, or the predictive-variance $w^\top S_i w$) raises out-of-sample AUC. *(iv) Misspecification stress test.* We repeat the coverage protocol but generate data from four perturbed processes — residuals sharing a nuisance factor (correlation $\rho \in \{0, .15, .30, .45\}$), an added ninth latent factor loading a random item subset (strength $\in \{0, .25, .5, .75\}$), loadings multiplicatively perturbed ($\lambda_{jk}(1 + \varepsilon\xi)$, $\varepsilon \in \{0, .10, .20, .30\}$), and Student- t residuals ($\text{df} \in \{\infty, 8, 5, 3\}$) — score each with the *unperturbed* projection, and track empirical 95% coverage against perturbation magnitude. Full specifications and seeds are in the analysis scripts released with the paper.

3 Results

We demonstrate the five instrument properties in the order a user meets them, on the fitted eight-axis map of $N = 9,013$ patients. A single real patient (§3.1) makes the whole chain concrete; the remaining subsections establish each property at the level of the full sample. Throughout, where a specific axis or patient appears it illustrates a measurement property, not a clinical claim.

3.1 Projection: a coordinate with its uncertainty, per patient

Figure 1 follows one patient from the bipolar cohort through the instrument. Projecting their observed cells through (7) returns a posterior coordinate on all eight axes together with its uncertainty (Fig. 1A). The two are inseparable and they are heterogeneous across axes for the same patient: the immunometabolic coordinate is pinned down to a posterior standard deviation of 0.22 by six observed indicators, whereas cognition and suicidality — on which this patient has no observed item — return the prior, a standard deviation near 0.99. The map does not paper over the gap with a point estimate; it reports a wide interval and labels the axis prior-dominated. This is the instrument’s first and most basic honesty: a coordinate is always accompanied by how much data produced it.

The same posterior supports three immediate reads. The patient’s archetype mixture (Fig. 1B) places 86% of their weight on a single corner of the simplex, so the eight-number coordinate compresses to an interpretable one-line summary without discarding the minority mixture. The interval-valued prognosis (Fig. 1C) is read off the coordinate’s neighbourhood, and — because the coordinate carries S_i — it too is an interval, not a point. And the value-of-information panel (Fig. 1D) names the single most informative next measurement for this patient: one verbal-memory test would lift the cognition axis from prior-dominated to a reliability of 0.65. Projection, summary, prognosis and next-measurement all read off one object.

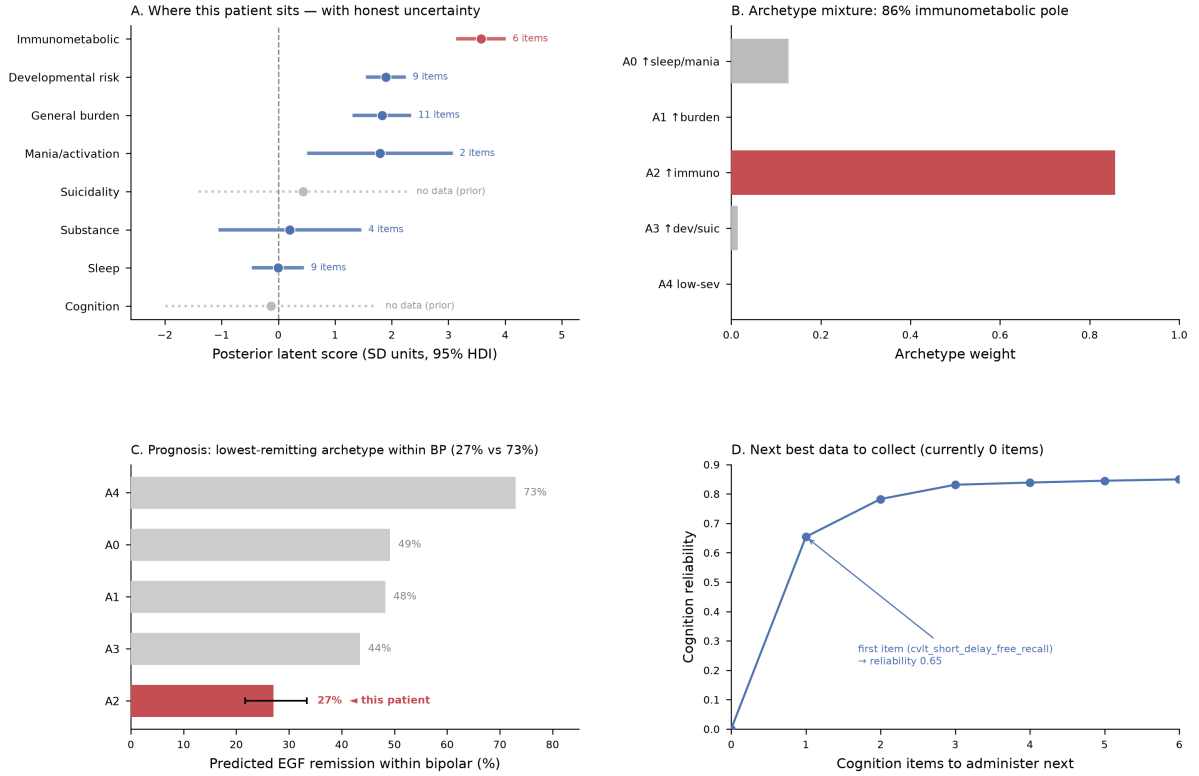


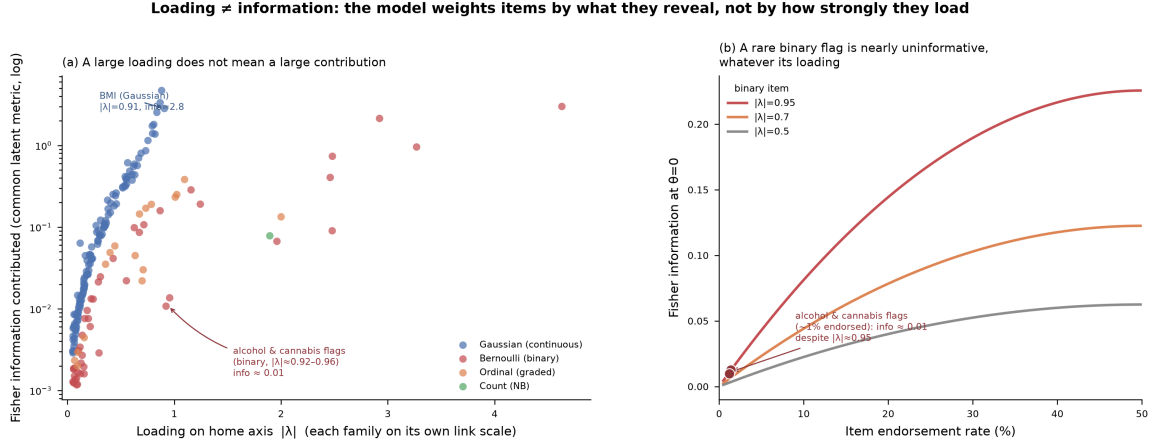
Figure 1: **Projection** — the instrument returns a coordinate, its uncertainty, a summary, a prognosis, and the next measurement, for one patient. A real patient from the bipolar cohort projected through the measurement model (Eq. 7). **(A)** Posterior coordinate on all eight axes with 95% credible intervals; axes with no observed indicator return the prior and are labelled as such, alongside the per-axis count of observed items. **(B)** The patient’s archetype-simplex membership, concentrating 86% of weight on one corner. **(C)** The interval-valued prognosis read from the coordinate’s neighbourhood. **(D)** Value of information: the single next measurement that most reduces this patient’s coordinate uncertainty, lifting the prior-dominated cognition axis to reliability 0.65. The panels are one object read four ways.

3.2 Information: what localizes a patient is not the loading

Reading the map as an instrument reorders the item bank. Figure 2 contrasts each indicator’s loading on its home axis with the Fisher information it actually contributes under its own likelihood family (9). The two diverge sharply. For a binary indicator the information is $\lambda^2 p(1-p)$, so a symptom flag endorsed by a few percent of patients contributes almost nothing to localization however strongly it loads: at 30% endorsement a unit-loading item carries information ≈ 0.10 , at 50% it carries 0.33, and only a near-ubiquitous item approaches the information a single well-scaled continuous measure supplies. The practical consequence is a target-driven item count: to reach a posterior standard deviation of 0.5 on an axis requires accumulated information $\gtrsim 3$, which is roughly four items at loading 0.7 but ~ 9 at loading 0.5 and ~ 31 at loading 0.3. What localizes a patient is the information an item carries, and information — not loading — is the currency the rest of the instrument spends.

3.3 Adaptive sampling: a tight upper bound, and where the bank is the limit

Accumulating information most-informative-first yields the reliability-versus-items curve for each axis (Fig. 3A). A small number of well-chosen indicators recovers most of an axis’s reliable signal: three to six items reach the useful range on the well-populated axes. Two axes cannot



(a) Each point is one indicator: its loading on its home axis (on that likelihood family's natural link scale, so cross-family magnitudes are not directly comparable) versus the Fisher information it contributes to that axis score in the common latent metric — the quantity that IS comparable, computed exactly per family at the population mean. Continuous markers with moderate loadings dominate; two substance flags load near 0.95 yet contribute ~ 0.01 , a hundred-fold less than a mid-loading continuous item. (b) For a binary item, information is $Np(1-p)$; at $\sim 3\%$ endorsement it collapses regardless of loading. This is why the substance axis, nominally carried by two high-loading flags, is in fact near-empty — and why an information-based reading, not a loading-based one, is the correct lens on a mixed-type measurement model.

Figure 2: Information, not loading, localizes a patient. For every modelled indicator, its loading on its home axis versus the Fisher information it contributes about that axis under its own likelihood family (Eq. 9). Continuous items convert loading into information efficiently; binary flags carry information $\lambda^2 p(1-p)$ that collapses at low endorsement, so a strongly-loading rare symptom localizes almost no one. The right panel translates this into a target-driven item count: reaching a posterior standard deviation of 0.5 needs ~ 4 items at loading 0.7 but ~ 31 at loading 0.3.

be rescued by more items of the same kind — mania/activation plateaus at reliability 0.41 (a 2-item bank) and substance at 0.43 (a 4-item bank). These ceilings are a property of the *instrument content*, not of missingness: no adaptive schedule and no completed data matrix can exceed them, because the information simply is not in the bank. Naming them is a service the instrument performs that a point-estimate map cannot.

The fixed most-informative-first order is the population-mean curve, an upper bound on any patient-adaptive schedule. Simulating a full CAT (§2.5) shows the bound is *tight* (Fig. 3B): on every axis whose leading items are Gaussian — whose information is θ -independent — patient-adaptive selection coincides with the fixed order to within 2×10^{-4} in reliability, because there is nothing to adapt to. Only the two axes with Bernoulli-topped banks show any adaptive signature at all, and it is small: suicidality diverges by at most 0.012 (at two items) and substance by 0.0015 at its four-item cap. The operational message is counter-intuitive but useful: for this instrument a fixed, pre-specified battery is essentially as efficient as a fully adaptive one, so the minimal-indicator rule can be published as a static list rather than requiring adaptive delivery infrastructure.

3.4 Value of information: designing an efficient shared battery

The closed-form entropy reduction (10) lets a greedy rule assemble a battery that maximises information per item across all eight axes at once. Figure 4 shows the cross-axis reliability rising with battery size; a shared 27-item battery reaches a mean reliability of 0.70 across the eight axes, a substantial reduction from administering every bank in full. The same accounting, applied within each cohort, localizes where an axis is under-measured: schizophrenia patients carry far fewer immunometabolic observations than bipolar patients (20.7 versus 34.9 items on average) and fewer sleep observations (3.9 versus 8.4), so the value of adding those measurements is concentrated in the schizophrenia cohort. Value of information thus turns the map into a

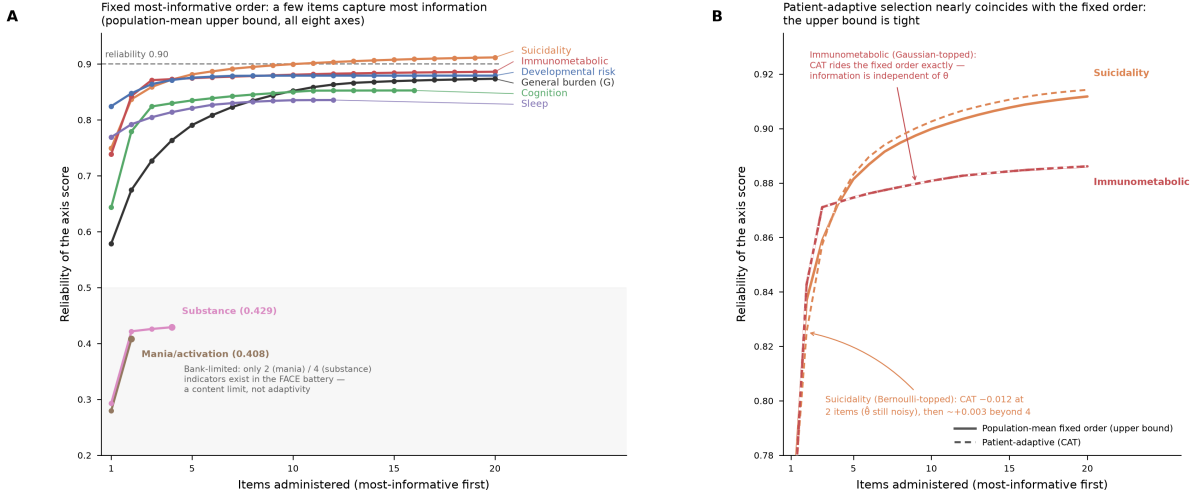


Figure 3: Adaptive assessment — the fixed order is a tight upper bound, and two axes are bank-limited. (A) Reliability versus number of items administered, most-informative-first, for all eight axes; three to six items reach the useful range on well-populated axes, while mania/activation (reliability 0.41, 2-item bank) and substance (0.43, 4-item bank) are capped by instrument content, not missingness. (B) A simulated computerized adaptive test (500 synthetic patients per axis, seed 20240704) versus the fixed population-mean order. On Gaussian-topped axes (e.g. immunometabolic) the two coincide, because θ -independent information leaves nothing to adapt to; only Bernoulli-topped axes (e.g. suicidality) show a small adaptive signature. The population-mean reliability curve is therefore a tight upper bound on adaptive efficiency.

design tool — it prescribes not just how few items suffice in general, but which cohort should be measured more, and on what.

3.5 Trust I — the coordinate’s uncertainty is calibrated

An instrument that reports intervals is only useful if the intervals are honest. Simulating 10,000 patients with known coordinates and real missingness patterns (§2.7), the credible intervals are well-calibrated across the board (Fig. 5). Empirical coverage tracks nominal at every level — 0.501, 0.801, 0.900 and 0.949 against nominal 0.50, 0.80, 0.90 and 0.95 — both averaged over the eight axes and for the joint eight-dimensional credible ellipsoid (0.947 at the 95% level; Monte-Carlo standard error 0.0022). Coverage is flat as a function of how much data the patient supplied (Fig. 5B): from prior-dominated axes with zero observed items, through one or two, up to eleven or more, the 95% interval covers at 0.947–0.952, every bin within its binomial error of nominal. The zero-item bin recovers nominal coverage by construction — the projection returns the prior when there is no data, and the test confirms the machinery does exactly that. The map’s uncertainty is therefore not decorative: an interval labelled 95% contains the truth 95% of the time, whether the patient was richly or barely assessed.

3.6 Trust II — the archetype summary reconstructs the coordinate faithfully

The population is a continuum, not a set of types, and this dictates the summary. The best five-way partition of the real coordinate cloud separates patients no better than a structureless Gaussian null (silhouette 0.140 versus 0.137 ± 0.002 ; $z = 1.13$, not significant), and most patients sit in the interior of the simplex as blends rather than at a corner (Fig. 6B): the median archetype-weight entropy is well away from zero, and the majority have no single dominant corner. When there are no natural clusters, a hard-partition “biotype” claim is unjustified, and

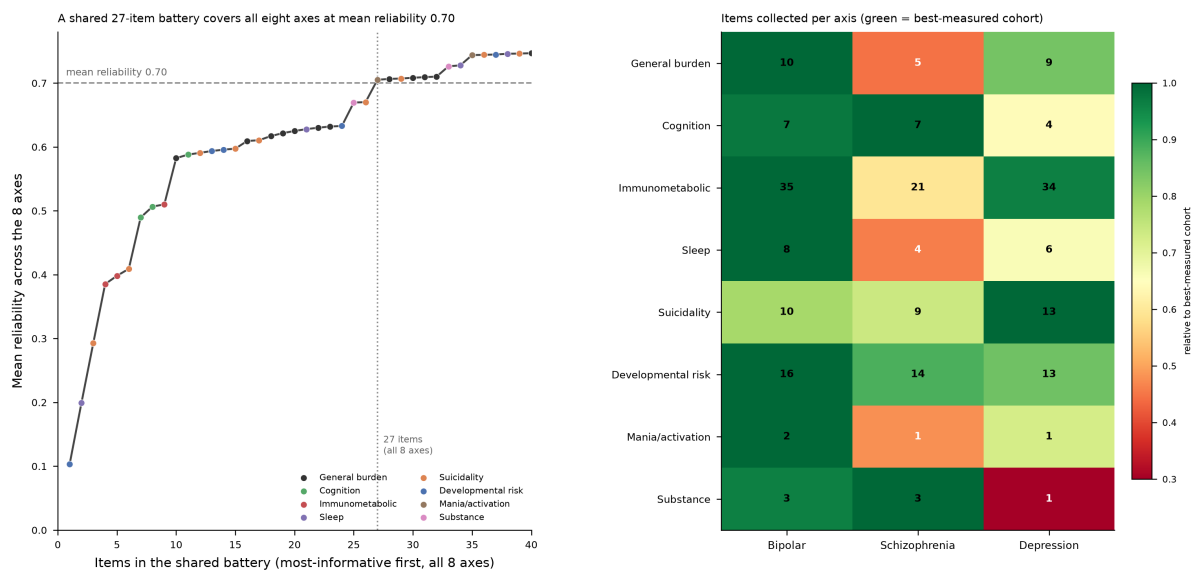


Figure 4: Value of information — designing an efficient shared battery and locating under-measured cohorts. The closed-form entropy reduction from each candidate item (Eq. 10) drives a greedy cross-axis selection: mean reliability across the eight axes rises with battery size, reaching 0.70 for a shared 27-item battery. Applied per cohort, the same accounting shows where an axis is under-measured (e.g. schizophrenia carries 20.7 immunometabolic items on average versus 34.9 in bipolar disorder), concentrating the value of additional measurement.

the honest low-dimensional summary is not a set of groups but a set of *reference poles* that each patient is a graded mixture of — exactly what the five-corner archetypal simplex provides (Fig. 6A), and exactly what neither a k -means partition nor a principal-component projection can.

Reconstruction fidelity is the wrong yardstick for this summary, and it is instructive to see why. Reconstructing each patient from their five archetype weights recovers 59.0% of the pooled coordinate variance (Fig. 6C). Taken alone that looks modest against a five-component principal-component projection, the unconstrained linear optimum at the same dimension, which recovers 79.6%. But the map’s coordinate variance is nearly isotropic — the leading principal component carries only 26.9%, and five components are needed to reach 80% — so variance capture is a weak test of any five-dimensional summary here. The expected variance a *random* five-dimensional subspace retains is exactly $k/d = 5/8 = 0.625$ regardless of the covariance spectrum, and a 500-seed simulation confirms it (0.624 ± 0.042). The archetype simplex therefore does not merely trail the linear optimum; it sits at the 21st percentile of *random* five-dimensional projections — slightly below what an arbitrary subspace achieves. Reconstruction R^2 cannot be the reason to prefer the simplex, because a coin-flip subspace beats it, and the fidelity rises only slowly with the number of corners ($R^2 \approx 0.53, 0.62, 0.69$ at four, five and six archetypes in a fresh refit whose five-corner value reproduces the production 0.59 to within optimizer reinitialization noise; Fig. 8B inset). What the simplex provides is what none of these linear or random summaries do: convex, non-negative, sum-to-one membership in five *named* clinical poles (Fig. 7) — a principal component is a signed contrast that mixes axes and that no patient can *be*, whereas an archetype corner is a near-real extreme profile a patient can be 70% of, with the attendant clinical reading. The categorical resolution bears this out: the archetype assignment accounts for $\eta^2 = 0.256$ of the variance in the map’s coordinates, against 0.026 for the DSM-5 diagnosis and 0.018 for cohort — an ≈ 9.7 -fold tighter description of where a patient sits, on the map’s own axes.

Read this way, the per-axis reconstruction pattern is a diagnostic rather than a scorecard:

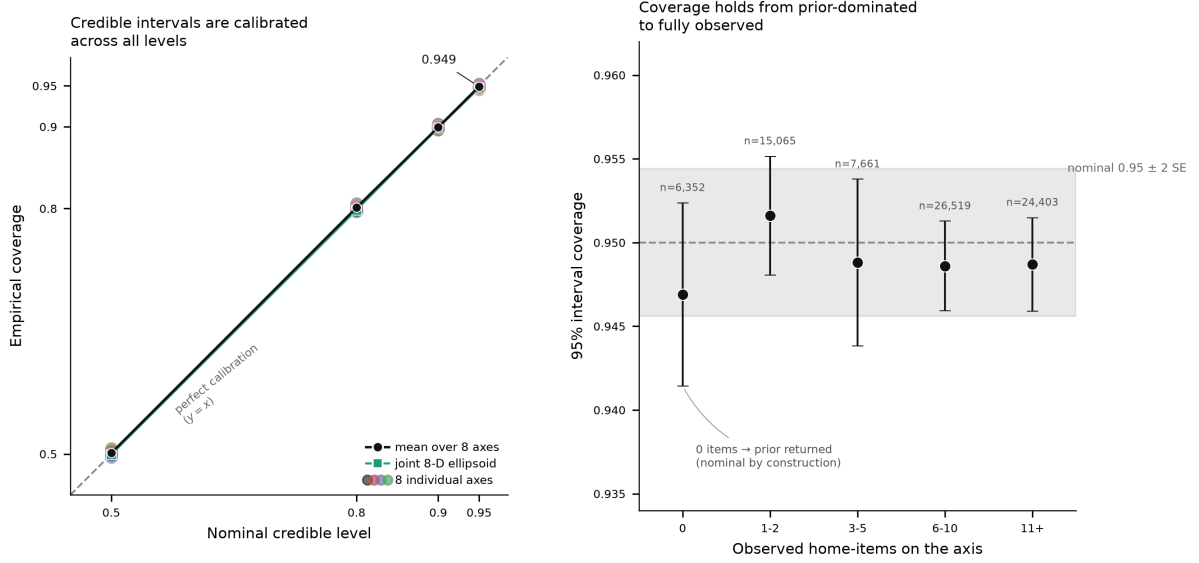


Figure 5: **Trust I — the posterior coordinate is calibrated.** From 10,000 synthetic patients with known coordinates and real bootstrapped missingness patterns (seed 20260704), projected through Fisher-scoring EAP (§2.7). **(A)** Empirical versus nominal credible level: the per-axis mean and the joint eight-dimensional ellipsoid both fall on the identity line (0.949 at nominal 0.95). **(B)** 95% interval coverage stratified by the number of observed indicators on the axis, with per-bin binomial error bars; coverage is flat from prior-dominated (0 items, nominal by construction) to richly observed (11+).

the simplex is faithful where the corners are well separated (immunometabolic $R^2 = 0.89$, overall severity 0.86, sleep 0.76, developmental risk 0.75) and near-flat where they are not (cognition 0.06, substance ≈ 0) — the same two bank-limited axes the information analysis already flagged, now visible as axes along which no five poles could separate patients because the items carry too little information. And reconstruction does not degrade for the ambiguous “fog” patients: error correlates *negatively* with archetype entropy (Fig. 6D; $r = -0.13$, $p < 10^{-30}$), because interior blends lie inside the corner hull whereas near-corner patients include hull-exceeding extremes. The simplex is thus the defensible summary of a continuum with no clusters: interpretable poles, graded membership, and a fidelity profile that reports — rather than hides — the axes on which the instrument itself is weak.

3.7 Utility I — the information rule changes which items you would collect

Two of the instrument’s claims — that information, not loading, should rank items, and that the archetype simplex is worth its fidelity cost — can be tested as head-to-head design decisions rather than asserted. Both hold, and both matter (Fig. 8). Assembling the shared battery by the information rule of §3.4 against the naive alternative of ranking items by loading magnitude, at the same 27-item budget, yields mean reliability 0.705 for the information rule against 0.599 for the loading rule — a 0.106 gap across the eight axes (Fig. 8A). The loading rule spends its early picks on high-loading but low-prevalence indicators that localize almost no one, and does not catch up until roughly item sixteen. Choosing items by the information they carry is thus not a refinement of choosing by loading; it selects a materially different, and measurably better, battery.

The archetype summary invites the opposite test — whether its interpretability is bought at an indefensible fidelity cost — and the answer reframes what the 59% reconstruction figure means (Fig. 8B). Anchored against the full range, the archetype simplex (0.59) sits below not only the

five-component linear optimum (0.80) but a *random* five-dimensional subspace (0.62), because the near-isotropic coordinate variance makes any five-dimensional projection capture roughly $k/d = 0.625$. Reconstruction R^2 cannot adjudicate between the summaries, and the choice is therefore made on what only the simplex delivers — named, convex, per-patient membership in five clinical poles — not on variance retained.

3.8 Utility II — carrying the posterior improves downstream prognosis, and calibration is robust

The instrument’s foundational promise is that a coordinate carried with its uncertainty is worth more than a point estimate. Simulating item responses from the fitted model with a known latent-to-remission rule, and comparing the full expected-a-posteriori coordinate against a naive per-axis sum-score of the same observed items, the posterior predicts the outcome better at every level of item completeness, and the advantage grows as data thin (Fig. 9A): the out-of-sample AUC gap is +0.018 on complete data and +0.039 at a quarter of items observed, against an oracle AUC of 0.799 on the true latent. The mechanism is the posterior’s shrinkage — where a sum-score propagates the full noise of a sparsely-measured axis, the projection pulls it toward the prior in proportion to how little was observed — which is precisely why the gain concentrates in the sparse regime. The effect is not a simulation artifact: over twelve seeds and an alternative outcome rule the posterior wins every time (gap $+0.019 \pm 0.003$ on full data, $+0.032 \pm 0.006$ when sparse). We note the boundary of this claim honestly: on the real cohort, triaging patients by their posterior uncertainty does not raise remission prediction (out-of-sample AUC 0.709 whether or not the least-certain patients are set aside), because remission at follow-up is driven substantially by treatment and course factors the baseline coordinate cannot see; the demonstrated value of the posterior is the sharper coordinate, not a confidence-triage of who is predictable.

Because the calibration of §3.5 samples from the model’s own parameters, it establishes internal honesty but not robustness to misspecification. We therefore generate data from four data-generating processes that each violate a model assumption — correlated residuals (breaking local independence), an omitted ninth latent factor, multiplicatively mismatched loadings, and heavy-tailed (Student- t) residuals — and score them with the *unperturbed* projection (Fig. 9B). Coverage of the nominal 95% interval degrades gracefully in every case and collapses in none. Heavy tails are absorbed almost without loss (coverage 0.955 at t with three degrees of freedom); an omitted factor and mismatched loadings sag gently to 0.93–0.94; the steepest and most consequential violation, correlated residuals, drifts monotonically to 0.889 at an extreme shared-*nuisance* loading — the expected direction, since local dependence is where factor-model intervals are most optimistic, yet even there the interval loses only six points of coverage. Calibration under this instrument is thus not a by-construction artifact: the intervals are honest when the model holds and fail predictably, not catastrophically, when it does not, with local dependence flagged as the assumption most worth checking on real data.

4 Discussion

We have argued that a transdiagnostic dimensional map is most useful when read not as a table of coordinates but as a measurement instrument, and we have shown that a single missingness-native Bayesian bifactor model makes five instrument properties computable and testable on one object: it projects each patient to a coordinate with its own uncertainty, weighs each indicator by the information it contributes, prescribes how few items suffice and where the item bank itself is the limit, values the cheapest next measurement, and — as we verified by simulation — reports uncertainty that is calibrated and a low-dimensional summary whose fidelity is quantified. None of the individual components is new: expected-a-posteriori scoring, Fisher information, adaptive

testing, value-of-information design, full-information treatment of missing data, and archetypal analysis are each standard. The contribution is their integration, and their joint validation on a real, pervasively incomplete, mixed-type transdiagnostic assessment at $N = 9,013$ — the setting where each component is usually assumed rather than demonstrated.

The instrument framing changes what a map is for. A point-estimate map invites the reader to treat every coordinate as equally trustworthy and every axis as equally measured; the honest answer, visible in Figure 1, is that a patient may be pinned to a posterior standard deviation of 0.22 on one axis and floated in from the prior on another, and that the map should say so. Once the map reports information rather than only position, three practical capabilities follow directly. First, the item bank can be audited: Figure 2 shows that loading and information part company, so a battery assembled to maximise loadings can be dominated by rare flags that localize no one, and the fix — reorder by information — is immediate. Second, assessment can be made cheaper without guessing: the minimal-indicator rule and the value-of-information battery (Figs. 3, 4) turn “which items do we actually need” into a closed-form calculation, and the finding that a fixed most-informative-first order is a tight upper bound on adaptive efficiency means the saving can be shipped as a static list rather than requiring adaptive-testing infrastructure. Third, under-measured cohorts are localized rather than assumed: the same accounting shows exactly which axis in which cohort would benefit most from additional measurement.

These capabilities would be hollow if the instrument’s machinery did not beat the simpler tools a reader already has, so we tested it against them directly. Ranking battery items by information rather than loading is not a cosmetic preference: at a fixed 27-item budget it buys 0.11 of mean reliability (Fig. 8A), because loading and prevalence pull in different directions. Carrying the full posterior coordinate rather than a per-axis sum-score improves out-of-sample prognosis, and does so most where it matters — under sparse measurement, the regime in which a point estimate is least trustworthy (Fig. 9A). And the calibration that a same-model simulation can only show to be internally consistent survives genuine misspecification: coverage degrades smoothly, never catastrophically, when the data-generating process breaks local independence, omits a factor, or violates Gaussianity (Fig. 9B). These are the comparisons a measurement venue asks for. On the two design decisions the instrument is put to — which items to collect, and which coordinate to carry downstream — it wins outright, not by a wide margin on any single one but consistently, and for mechanistic reasons the framework predicts in advance. On the third, reconstruction fidelity, it does not win and does not need to: we show the metric itself cannot separate a convex summary from a random subspace on this near-isotropic cloud, so the archetype simplex earns its place on interpretability and graded named membership rather than on variance retained. Winning where the comparison is decision-relevant, and dismantling the yardstick where it is not, is the honest reading of the four analyses.

Two results are worth stating as limits the instrument imposes on itself. The bank-limited axes — mania/activation and substance, capped at reliability 0.41 and 0.43 — cannot be improved by any amount of adaptive scheduling or data completion, because the information is not present in the items; the same two axes are the ones the archetype summary reconstructs worst (Fig. 6C). This is not a failure of the method but a diagnosis it delivers: these constructs need better instruments, not better analysis. And the five-corner convex summary retains 59% of coordinate variance against an 80% linear optimum — but the honest reading of that gap is not that the simplex is a worse projection. The map’s coordinate variance is nearly isotropic, so a *random* five-dimensional subspace already retains 62% (the analytic k/d), and the archetype simplex sits slightly below it: reconstruction R^2 is simply the wrong axis on which to judge this summary, because a coin-flip subspace beats it and a principal component — a signed contrast no patient can *be* — beats it further while providing nothing a clinician can act on. The simplex earns its place not by variance retained but by what it uniquely supplies: convex, named, per-patient membership in five clinical poles over a population that has no natural clusters to begin with (silhouette $z = 1.13$). Stating this plainly, rather than quoting the R^2 one would

prefer, is itself the point: an instrument that reports the axis on which it should *not* be judged is more useful than one that claims a fidelity victory it does not have.

Several limitations bound the claims. Calibration is established in simulation under the model’s own generative assumptions and real missingness patterns; it demonstrates that the projection machinery is internally honest — that its 95% intervals cover at 95% — but not that the fitted model is the true data generating process, which no coverage study can. Posterior standard deviations run slightly below residual error on the discrete-informed axes (a light-tailed Laplace approximation), so interval width is marginally optimistic even though the 95%-level coverage, set by the interval quantiles, is exact. The value-of-information accounting is greedy and Gaussian-entropy based, an upper bound on a jointly-optimal design. The reliabilities and item counts are properties of this bank in these cohorts and will differ for another assessment — which is precisely why the instrument is designed to be re-run on a new one. Finally, the adaptive-testing analysis is a single-axis simulation on model-generated responses; extending it to the joint eight-axis posterior with cross-axis information sharing is a natural next step that would only tighten the case for a fixed battery.

This paper deliberately reports no clinical or biological finding. The immunometabolic axis that recurs throughout as a worked example, the archetype corners, the cohort-level prognosis — these are substantive results, but they belong to the companion paper, where they are established on their own terms with the appropriate clinical framing and controls. The division is not cosmetic: a reader who wants to know whether the immunometabolic axis predicts remission should not have to adjudicate a measurement-model derivation to get there, and a reader who wants to reuse the instrument on a new cohort should not have to extract it from a clinical narrative. Kept separate, each paper is self-contained; the measurement instrument described here is what makes the clinical map trustworthy, and the clinical map is what makes the instrument worth building.

Relation to the companion paper

This paper describes the measurement instrument. The clinical and biological findings the instrument supports — the transdiagnostic map’s structure, the immunometabolic axis, the archetype-level prognosis — are reported in a companion paper (in preparation). The two are designed to be read independently: the present paper establishes that the map’s coordinates, uncertainties and summaries are trustworthy and efficiently obtainable; the companion paper interprets what they reveal about severe mental illness.

Data availability

The FACE database per-patient cohort data are confidential and governed by FondaMental’s data-access procedures, to which qualified researchers may apply. No per-patient data are reproduced in this article. Aggregate parameters, derived structure, the fitted loading and information tables, and all figure source values are available in the project’s regenerated aggregate outputs. The calibration and adaptive-testing analyses use only the binary observed/missing pattern of the real data (never any clinical value) together with model-generated synthetic responses.

Code availability

The full analysis pipeline (data harmonization, the Bayesian measurement engine, projection and scoring, the information/value-of-information accounting, and archetypal analysis), the harmonized variable dictionary, and configuration files are available at <https://github.com/>

AndriiKulakovskiy/face-trans-diagnostic-phenotypes.git. The scripts that regenerate every figure in this paper, and the new patient-adaptive, calibration and reconstruction analyses, are included. Every reported quantity is reproducible from the numbered pipeline; per-patient data are never required to regenerate the aggregate results and figures.

Ethics

The FACE cohorts were approved by the relevant institutional review boards and all participants provided written informed consent. [TBD: insert exact IRB/ethics committee names, approval numbers, and cohort registration identifiers.]

Competing interests

None declared.

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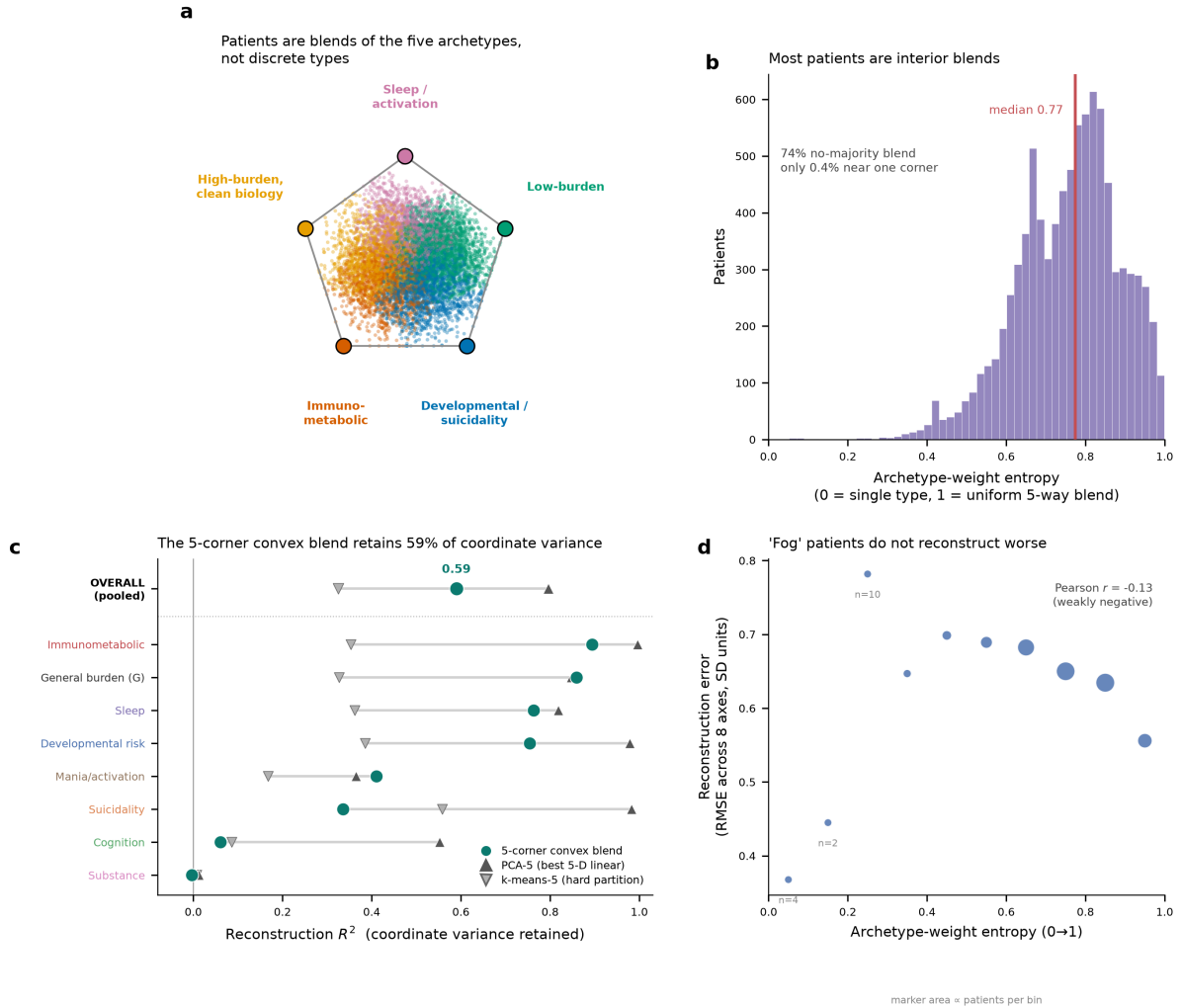


Figure 6: Trust II — the archetype simplex summarizes the continuum, and its reconstruction fidelity is quantified. (A) The five-corner archetypal simplex (Eq. 11) with all 9,013 patients projected into it, coloured by dominant corner; the interior is densely populated. (B) Distribution of archetype-weight entropy: most patients are blends, not types. (C) Reconstruction fidelity per axis — the five-corner convex blend ($R^2 = 0.59$ pooled) against a five-component PCA linear optimum (0.80) and a five-centroid hard partition (0.33); faithful where corners separate (immunometabolic 0.89), weak on the bank-limited axes (cognition, substance). (D) Reconstruction error versus archetype entropy: ambiguous “fog” patients do not reconstruct worse ($r = -0.13$).

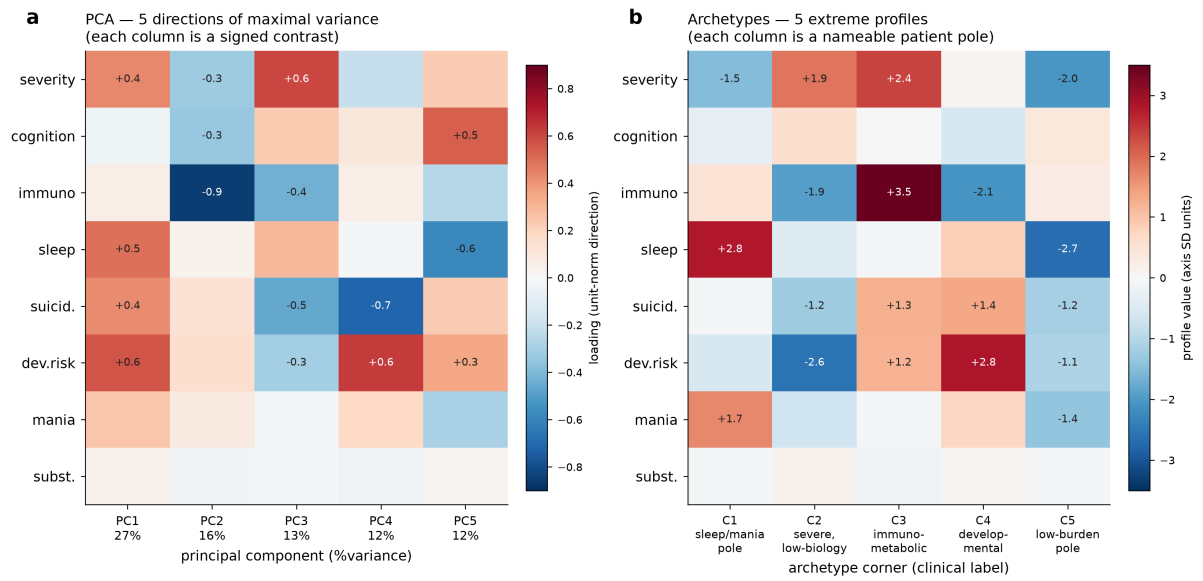


Figure 7: **What each low-dimensional summary produces on the map.** (A) The five leading principal components: each is a signed *contrast* that mixes axes (PC1 loads jointly on developmental risk, sleep, severity and suicidality; PC2 is essentially $-\text{immunometabolic}$), captures a named fraction of variance, and describes a *direction* no patient can be. (B) The five archetype corners: each is a near-real extreme *profile* on the eight axes with a clinical label (C3 = immunometabolic +3.5, severity +2.4; C1 = sleep +2.8, mania +1.7), so a patient can be described as a convex blend — “70% C3” — with an attendant clinical reading. Both summarize the same 8-dimensional cloud in five numbers; only the archetype corners are nameable poles.

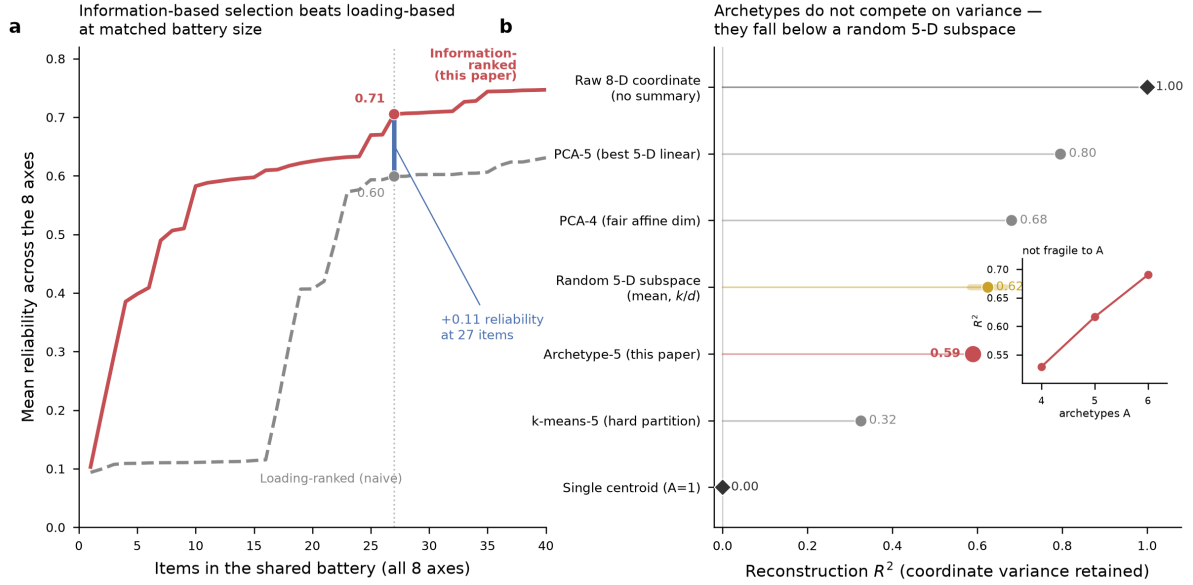


Figure 8: **The instrument’s choices change decisions.** (A) Building the shared battery by ranking items on Fisher information (this paper) versus on loading magnitude, over all eight axes: at the 27-item battery the information-ranked rule reaches mean reliability 0.71 against 0.60 for the loading-ranked rule (a +0.11 gap), because high-loading, low-prevalence items localize no one. (B) Reconstruction R^2 of the five-corner archetype simplex placed against the full $0 \rightarrow 1$ range: the raw eight-dimensional coordinate (1.00), the best five-dimensional linear projection (PCA-5, 0.80), a *random* five-dimensional subspace (mean 0.62, the analytic k/d value, 500 seeds), the archetype simplex (0.59), a five-centroid hard partition (0.33) and a single centroid (0.00). Because the coordinate variance is nearly isotropic, a random subspace retains *more* variance than the archetypes; R^2 is therefore uninformative here, and the case for the simplex rests on interpretability, not fidelity. Inset: reconstruction R^2 is stable across the number of corners.

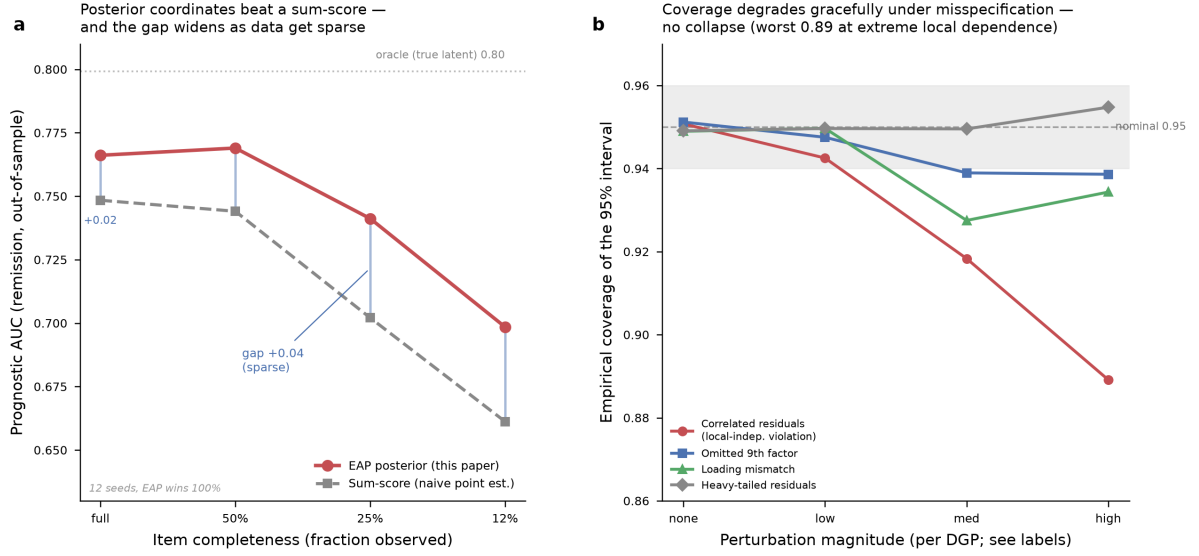


Figure 9: **The posterior earns its keep, and its intervals degrade gracefully.** (A) Head-to-head prognosis on model-simulated data with a known remission rule: the full expected-a-posteriori coordinate (this paper) against a naive per-axis sum-score, as item completeness falls. The posterior wins at every level and the gap *widens* as data get sparse (+0.02 at full data, +0.04 at 25% observed), because it shrinks noisy inputs toward the prior instead of averaging them blindly; over 12 seeds the posterior wins 100% of the time. The dotted line is the oracle AUC on the true latent (0.80). (B) Coverage of the nominal 95% interval when the data-generating process *violates* a model assumption, scored with the unperturbed projection: correlated residuals (local-independence violation), an omitted ninth factor, mismatched loadings, and heavy-tailed residuals. Coverage degrades smoothly and never collapses — the worst case, an extreme local-dependence violation, still covers at 0.89, and heavy tails are tolerated essentially without loss.

Supplementary Material

A transdiagnostic dimensional map as a self-calibrating measurement instrument: projection, information, and value of information under structured missingness

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This document gives the full derivations that the Methods section of the main text (§2.1–§2.7) states as results and defers here. It is self-contained: every object it uses is defined below, and cross-references of the form “Eq. (7)” point to the numbered equations of the main manuscript. Sections S1 and S2 develop the measurement model and the projection machinery from first principles; Sections S3–S7 derive in full the five instrument properties that are the paper’s contribution — the loading as a measurement slope, the information accounting and per-family Fisher information, adaptive sampling and value of information, the archetypal summary and its reconstruction fidelity, and the calibration protocol. Propositions are proved; everything reported in the main text is an evaluation of the objects defined here.

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S1 The measurement model from first principles

S1.1 Generative factor model and local independence

Index patients $i = 1, \dots, N$, indicators $j = 1, \dots, J$, specific factors $k = 1, \dots, K$ and one general factor G . Each patient carries a factor vector $f_i = (G_i, D_{i1}, \dots, D_{iK})^\top \in \mathbb{R}^{K+1}$ with

$$f_i \sim \mathcal{N}_{K+1}(\mathbf{0}, \Phi), \quad \text{diag}(\Phi) = \mathbf{1}, \quad (\text{S1})$$

the factor *correlation* matrix Φ carrying a unit diagonal to fix the latent scale. Each indicator is a linear reflection of the factors plus item-specific noise through a single linear predictor,

$$\eta_{ij} = \alpha_j + \lambda_{jG} G_i + \sum_{k=1}^K \lambda_{jk} D_{ik} + \beta_j^\top c_i, \quad (\text{S2})$$

with item intercept α_j , general loading λ_{jG} , specific loadings λ_{jk} (row j of the loading matrix $\Lambda = [\lambda_G \mid \Lambda_D] \in \mathbb{R}^{J \times (K+1)}$), and item-level covariates c_i that adjust an indicator’s mean but are never factor indicators. This is Eq. (1) of the main text. Stacking one patient’s items,

$$x_i = \mu_i + \lambda_G G_i + \Lambda_D D_i + \varepsilon_i, \quad \mu_i = \alpha + B c_i, \quad \varepsilon_i \sim \mathcal{N}(\mathbf{0}, \Psi), \quad (\text{S3})$$

with diagonal residual covariance $\Psi = \text{diag}(\sigma_1^2, \dots, \sigma_J^2)$.

Local independence. Given a patient’s latent scores the items are conditionally independent,

$$p(x_i \mid f_i, \Theta) = \prod_{j \in O_i} p(x_{ij} \mid f_i, \Theta), \quad (\text{S4})$$

i.e. the factors are the only reason items covary. In the Gaussian linear model this is identical to “ Ψ is diagonal”: local independence and diagonal residual covariance are two faces of one assumption. It is also why filling missing cells is forbidden — an invented value injects covariance the model would mistake for a factor. Because f_i is unobserved we integrate it out and multiply across independent patients,

$$L(\Theta) = \prod_{i=1}^N \int \left[\prod_{j \in O_i} p(x_{ij} \mid f_i, \Theta) \right] \mathcal{N}(f_i; \mathbf{0}, \Phi) \mathrm{d}f_i. \quad (\text{S5})$$

Equation (S2) is a bifactor exploratory structural-equation model: every indicator may load on G and on its home specific factor, while off-home cells are governed by a sparsity-inducing prior (§S1.4) that keeps the map mostly simple-structure yet lets a few evidence-backed cross-loadings emerge. The fitted map has $K+1 = 8$ factors: G (overall burden) and seven specific axes — cognition, immunometabolic, sleep, mania/activation, suicidality, developmental-risk and substance. The fitted map carries 109 indicators (88 in the Gaussian-copula continuous block, 21 in the explicit non-Gaussian block).

S1.2 Mixed-likelihood observation model and the Gaussian copula

Coercing variables onto a common scale would manufacture covariance and bias the very factors being estimated. Each indicator instead keeps the observation density appropriate to its type, all sharing the linear predictor (S2):

$$\text{continuous / } z\text{-score: } X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2), \quad (\text{S6})$$

$$\text{skewed laboratory: } \log X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2), \quad (\text{S7})$$

$$\text{ordinal } (C_j \text{ levels}): \Pr(X_{ij} \leq c) = \text{logit}^{-1}(\tau_{jc} - \eta_{ij}), \quad \tau_{j1} < \dots < \tau_{j,C_j-1}, \quad (\text{S8})$$

$$\text{binary: } X_{ij} \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_{ij})), \quad (\text{S9})$$

$$\text{count: } X_{ij} \sim \text{NegBinom}(\mu_{ij} = e^{\eta_{ij}}, r_j). \quad (\text{S10})$$

The reported map replaces the raw Gaussian likelihood on the continuous block with a semiparametric Gaussian-copula model: each continuous indicator is mapped to a latent Gaussian score by its frozen empirical distribution function $\hat{F}_j$, $z_{ij} = \Phi_{\mathcal{N}}^{-1}(\hat{F}_j(X_{ij}))$, and the factor model is fit on z_{ij} . Because the rank transform is monotone it alters none of the identification arguments below. Writing $F_j(\cdot)$ for indicator j 's density and $O_i \subseteq \{1, \dots, J\}$ for its observed indicators, the observed-data log-likelihood sums over observed cells only,

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^N \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j), \quad (\text{S11})$$

which is Eq. (3) of the main text: a missing cell contributes no term and no completed $N \times J$ matrix is ever formed.

S1.3 Missing data is the marginal property, not imputation

Proposition 1 (Marginalization). *For patient i with observed continuous set $C_i \subseteq O_i$, the model-implied distribution of the observed sub-vector is obtained by deleting the unobserved coordinates from (μ, Σ) :*

$$X_{i,C_i} \sim \mathcal{N}(\mu_{i,C_i}, \Sigma_{C_i}), \quad \Sigma_{C_i} = \Lambda_{C_i} \Phi \Lambda_{C_i}^\top + \Psi_{C_i}, \quad (\text{S12})$$

with Λ_{C_i} the observed-row loadings and $\Psi_{C_i} = \text{diag}(\sigma_j^2)_{j \in C_i}$.

Proof. A foundational property of the multivariate normal is that marginals are read off by deleting coordinates: if $x \sim \mathcal{N}(\mu, \Sigma)$ then any sub-vector x_C is $\mathcal{N}(\mu_C, \Sigma_{CC})$. Apply this to $x_i \sim \mathcal{N}(\mu_i, \Sigma)$ with $\Sigma = \Lambda \Phi \Lambda^\top + \Psi$ (Proposition 3); the submatrix Σ_{CC} retains exactly the observed rows and columns of Λ and the observed diagonal of Ψ , giving (S12). $\square$

Hence “integrating out” the missing cells is not an extra step but the marginal property itself; summing $\log \mathcal{N}(\cdot)$ over each patient’s observed sub-vector is precisely the full-information maximum-likelihood (FIML) objective for incomplete multivariate-normal data. This is the formal justification of the no-imputation invariant (S11): any completed matrix would inject between-cell covariance that (S4) would misattribute to a latent factor.

S1.4 Identifiability: rotation invariance and how the prior resolves it

Proposition 2 (Rotation invariance of the likelihood). *For any invertible $M \in \mathbb{R}^{(K+1) \times (K+1)}$, the reparameterization $\Lambda^* = \Lambda M$, $\Phi^* = M^{-1} \Phi M^{-\top}$, $f^* = M^{-1} f$ leaves the implied covariance, and hence the Gaussian likelihood, unchanged:*

$$\Lambda^* \Phi^* \Lambda^{*\top} + \Psi = \Lambda M (M^{-1} \Phi M^{-\top}) M^\top \Lambda^\top + \Psi = \Lambda \Phi \Lambda^\top + \Psi = \Sigma. \quad (\text{S13})$$

Proof. Immediate from $MM^{-1} = I$: the inner $M(M^{-1}\Phi M^{-\top})M^{\top}$ telescopes to Φ . Since x_i is determined in distribution by (μ_i, Σ) and Σ is invariant, so is the likelihood. $\square$

A factor model is therefore not identified from the likelihood alone. Confirmatory factor analysis breaks this by fixing loadings to exactly zero; the present model breaks it with the prior. Each (indicator, factor) cell is assigned a role and prior

$$\lambda_{jk} \mid \rho(j, k) \sim \begin{cases} \mathcal{N}_+(0.70, 0.25^2) & \text{PRIMARY (HOME CELL),} \\ \text{regularized horseshoe (S15)} & \text{PLAUSIBLE CROSS,} \\ \mathcal{N}(0, 0.05^2) & \text{UNLIKELY,} \\ \delta_0 & \text{FORBIDDEN,} \end{cases} \quad (\text{S14})$$

with $\mathcal{N}_+$ truncated to $(0, \infty)$ and δ_0 fixing the loading at zero. The likelihood is rotation-invariant but the prior is not: a rotation moves a loading the prior wanted near zero away from zero, lowering $p(\Lambda)$, so the posterior $p(\Lambda \mid X) \propto p(X \mid \Lambda) p(\Lambda)$ concentrates at the orientation where small loadings align with the soft-zero priors and home loadings are positive and substantial; the positive truncation removes the residual sign-flip. The plausible-cross cells use a regularized (“Finnish”) horseshoe,

$$\lambda_j = z_j \tau \tilde{\eta}_j, \quad \tilde{\eta}_j = \sqrt{\frac{c^2 \eta_j^2}{c^2 + \tau^2 \eta_j^2}}, \quad z_j \sim \mathcal{N}(0, 1), \quad \eta_j \sim \text{HalfCauchy}(1), \quad \tau \sim \text{HalfNormal}(\tau_0), \quad (\text{S15})$$

with $\tau_0 = 0.05$ and slab $c = 0.30$: default-off (global τ shrinks the set toward zero), evidence-on (heavy-tailed local η_j lets a genuine cross-loading escape), and magnitude-capped ($\tilde{\eta}_j \rightarrow c/\tau$, so an escaped loading stays small). This is the honest treatment of weak signal in a factor with few indicators.

Corollary 1 (Flat-prior confirmation). *Under a flat loading prior (identification constraints only) the posterior mode equals the maximum-likelihood (FIML) solution. A prior-free refit landing on the same loadings therefore demonstrates that the structure is recovered from the data, not manufactured by the priors; empirically this refit reproduced the map to Tucker congruence $\varphi = 1.00$.*

S1.5 The covariance identity and the correlated- G estimand

Proposition 3 (Bifactor covariance). *Under (S3) with $\text{Var}(G_i) = 1$, $\text{Cov}(D_i) = \Phi_D$, $\mathbb{E}[\varepsilon_i \varepsilon_i^{\top}] = \Psi$ diagonal, and the bifactor orthogonality $\mathbb{E}[G_i D_i^{\top}] = \mathbf{0}$ together with $(G_i, D_i) \perp \varepsilon_i$,*

$$\Sigma = \text{Cov}(x_i) = \underbrace{\lambda_G \lambda_G^{\top}}_{\text{general: rank one}} + \underbrace{\Lambda_D \Phi_D \Lambda_D^{\top}}_{\text{specific factors}} + \underbrace{\Psi}_{\text{private noise}} = \Lambda \Phi \Lambda^{\top} + \Psi. \quad (\text{S16})$$

Proof. Expand $\Sigma = \mathbb{E}[(x_i - \mu_i)(x_i - \mu_i)^{\top}]$ with $x_i - \mu_i = \lambda_G G_i + \Lambda_D D_i + \varepsilon_i$. The cross-terms $\lambda_G \mathbb{E}[G_i D_i^{\top}] \Lambda_D^{\top}$ and $\mathbb{E}[(\lambda_G G_i + \Lambda_D D_i) \varepsilon_i^{\top}]$ vanish by $\mathbb{E}[G_i D_i^{\top}] = \mathbf{0}$ and by independence of factors and noise respectively. The survivors are $\lambda_G \lambda_G^{\top} \text{Var}(G_i)$, $\Lambda_D \Phi_D \Lambda_D^{\top}$ and Ψ , giving (S16). $\square$

The general factor leaves a rank-one footprint: every pair of items covaries through G by exactly $\lambda_{jG} \lambda_{j'G}$. The correlated- G arm (Eq. (4) of the main text) unzips the zero off-diagonal block, freeing $\phi_{Gd} = \text{Corr}(G, D)$, whose immunometabolic entry near zero against far larger cognition and sleep entries is the estimand the companion paper interprets. Here it is used only as a worked example of the instrument.

S2 Projection: the Woodbury reduction and EAP scoring

S2.1 Woodbury reduction of the observed-cell covariance

Evaluating (S12) naively costs $\mathcal{O}(|C_i|^3)$ per patient. The Woodbury identity and matrix-determinant lemma reduce it to the $(K+1)$ -dimensional factor space.

Lemma 1 (Woodbury reduction). *With Ψ_{C_i} diagonal and Λ_{C_i} the observed-row loadings,*

$$\Sigma_{C_i}^{-1} = \Psi_{C_i}^{-1} - \Psi_{C_i}^{-1} \Lambda_{C_i} (\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i})^{-1} \Lambda_{C_i}^\top \Psi_{C_i}^{-1}, \quad (\text{S17})$$

$$\log |\Sigma_{C_i}| = \log |\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i}| + \log |\Phi| + \sum_{j \in C_i} \log \sigma_j^2. \quad (\text{S18})$$

Proof. Equation (S17) is the Woodbury matrix identity applied to $\Sigma_{C_i} = \Psi_{C_i} + \Lambda_{C_i} \Phi \Lambda_{C_i}^\top$ with the low-rank term $\Lambda_{C_i} \Phi \Lambda_{C_i}^\top$; (S18) is the matrix-determinant lemma $\log |\Psi + U \Phi U^\top| = \log |\Phi^{-1} + U^\top \Psi^{-1} U| + \log |\Phi| + \log |\Psi|$ with $U = \Lambda_{C_i}$ and Ψ diagonal. $\square$

The inner solve is $\mathcal{O}((K+1)^3)$, independent of $|C_i|$; patients sharing a missingness pattern share the inner $(K+1) \times (K+1)$ factorization, computed once per pattern. This makes full-sample ($N = 9,013$) projection tractable and is Eq. (6) of the main text; the non-Gaussian block retains explicit per-patient latents.

S2.2 EAP factor scores and Bayes/FIML equivalence

Proposition 4 (EAP scores). *For an all-continuous observed pattern, the posterior of f_i given the observed cells is Gaussian with mean and covariance*

$$\hat{f}_i = \mathbb{E}[f_i | X_{i,C_i}] = \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} (X_{i,C_i} - \mu_{i,C_i}), \quad S_i = \Phi - \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} \Lambda_{C_i} \Phi, \quad (\text{S19})$$

which is Eq. (7) of the main text.

Proof. Under (S1) and (S3) the joint of (f_i, X_{i,C_i}) is Gaussian with $\text{Cov}(f_i) = \Phi$, $\text{Cov}(X_{i,C_i}) = \Sigma_{C_i}$ and cross-covariance $\text{Cov}(f_i, X_{i,C_i}) = \Phi \Lambda_{C_i}^\top$. The Gaussian conditioning formula $\mathbb{E}[f | X] = \text{Cov}(f, X) \text{Cov}(X)^{-1} (X - \mu_X)$ and $\text{Cov}(f | X) = \text{Cov}(f) - \text{Cov}(f, X) \text{Cov}(X)^{-1} \text{Cov}(X, f)$ give (S19) directly. $\square$

Proposition 5 (Equivalent precision form). *The posterior covariance (S19) admits the precision form*

$$S_i^{-1} = \Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i}, \quad (\text{S20})$$

so the posterior precision is the prior precision plus a data term that is a sum of per-item contributions.

Proof. Substitute the Woodbury form (S17) into $S_i = \Phi - \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} \Lambda_{C_i} \Phi$ and simplify; the low-rank correction telescopes to give $S_i = (\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i})^{-1}$. Equivalently, (S20) is the standard Gaussian information form of the posterior of a linear model with prior precision Φ^{-1} and per-observation precision $\Psi_{C_i}^{-1}$ propagated through Λ_{C_i} . Inverting gives the covariance form (S19). $\square$

Equation (S20) is the pivot of the whole instrument: it exhibits the posterior precision as *additive over observed items*, which §S4 turns into the information accounting. When non-Gaussian cells are present the posterior is no longer exactly Gaussian; $\hat{f}_i$ and S_i are then the posterior mean and covariance obtained by MCMC or, for the calibration study, by the Fisher-scoring Laplace approximation of §S7, which reduces to (S19) exactly on all-Gaussian patterns.

Proposition 6 (Bayes/FIML equivalence). *The marginalized Bayesian observed-data model and FIML optimize the same observed-data likelihood (S11) and differ only by the priors (S14). In particular (Corollary 1) the flat-prior MAP equals the FIML estimate, so a standalone FIML estimator adds no independent evidence and confirmation is done in-engine.*

Proof. By Proposition 1 the per-patient marginal density is the incomplete-data multivariate-normal density on C_i , whose log summed over patients is the FIML objective. The posterior is this likelihood times the prior; with a flat prior the log-posterior equals the log-likelihood up to a constant, so its mode is the FIML maximizer. $\square$

Finally, the structure of (S19) makes the instrument’s honesty on unmeasured axes mechanical: on an axis with no observed indicator the corresponding rows of Λ_{C_i} are empty, the update term in (S20) vanishes on that coordinate, and the posterior returns the prior $\mathcal{N}(0, \Phi)$ — a wide interval, never a false-precise point. The per-patient posterior covariance S_i is exported and propagated into every downstream analysis; a patient with one observed indicator on a factor is not treated as equally characterized as one with six.

S3 The loading as a measurement slope

The main text (§2.2) separates two objects that (S2) places on the same line: the per-patient latent *position* f_i , and the fixed *slope* at which a measurement changes as that position moves. This section derives the slope on both the latent and the response scale.

Proposition 7 (Latent-scale slope). *On the linear-predictor scale the sensitivity of indicator j to factor k is exact, constant across patients, and equal to the loading:*

$$\frac{\partial \eta_{ij}}{\partial f_{ik}} = \lambda_{jk}. \quad (\text{S21})$$

Hence the loading matrix Λ is the Jacobian $\partial \eta_i / \partial f_i$ of the measurement with respect to the latent coordinate.

Proof. Differentiate (S2) with respect to f_{ik} ; the intercept, the covariate term $\beta_j^\top c_i$ and the other factor terms do not depend on f_{ik} , leaving λ_{jk} . $\square$

Proposition 8 (Response-scale slope). *For a single-index likelihood with conditional mean $\mathbb{E}[X_{ij} | f_i] = h_j(\eta_{ij})$ (link mean function h_j), the marginal effect of a factor on the indicator and its population average are*

$$\frac{\partial \mathbb{E}[X_{ij} | f_i]}{\partial f_{ik}} = h'_j(\eta_{ij}) \lambda_{jk}, \quad s_{jk} \equiv \mathbb{E} \left[\frac{\partial \mathbb{E}[X_{ij} | f_i]}{\partial f_{ik}} \right] = \kappa_j \lambda_{jk}, \quad \kappa_j = \mathbb{E} [h'_j(\eta_{ij})] \geq 0, \quad (\text{S22})$$

which is Eq. (5) of the main text.

Proof. The chain rule applied to $\mathbb{E}[X_{ij} | f_i] = h_j(\eta_{ij})$ with $\partial \eta_{ij} / \partial f_{ik} = \lambda_{jk}$ (Proposition 7) gives the first identity. Taking the expectation over the population distribution of η_{ij} and factoring out the patient-independent λ_{jk} defines $\kappa_j = \mathbb{E}[h'_j(\eta_{ij})]$, which is non-negative because each family’s mean function h_j is non-decreasing ($h'_j = \sigma^{-2}$ -scaled identity for the Gaussian, $p(1-p)$ for the logit-Bernoulli, e^η for the log-count). $\square$

The loading is therefore *how strongly an item reflects a factor*. It is not how much that item localizes a patient — that is the Fisher information of §S4, and the two coincide only for the Gaussian family, where h'_j is constant. The response-scale average κ_j is what makes a unit-loading rare binary flag a weak *measurement* despite a large λ : $\kappa_j = \mathbb{E}[p_{ij}(1-p_{ij})]$ collapses as the endorsement probability leaves $\frac{1}{2}$.

S4 Information accounting: what localizes a patient

S4.1 Additivity of posterior precision

Proposition 9 (Additive precision). *Under local independence (S4) the posterior precision of the coordinate is the prior precision plus a sum of per-item Fisher-information contributions,*

$$S_i^{-1} = \Phi^{-1} + \sum_{j \in O_i} \mathcal{I}_j(f_i), \quad \mathcal{I}_j(f_i) = -\mathbb{E}_{X_{ij}|f_i} \left[\nabla_{f_i}^2 \log F_j(X_{ij} | \eta_{ij}) \right], \quad (\text{S23})$$

which is Eq. (8) of the main text. Each observed cell adds a positive semidefinite term; nothing an item measures is ever subtracted.

Proof. The log-posterior of f_i given the observed cells is, up to a constant, $\log p(f_i | X_{i,O_i}) = \log \mathcal{N}(f_i; \mathbf{0}, \Phi) + \sum_{j \in O_i} \log F_j(X_{ij} | \eta_{ij})$, the sum over items following from local independence (S4). The negative Hessian in f_i of the prior term is Φ^{-1} ; the negative expected Hessian of each item term is by definition the item's Fisher information $\mathcal{I}_j(f_i)$ about the factors. Summing gives (S23); each $\mathcal{I}_j$ is positive semidefinite as an expected outer product of score vectors (below), so observing an item can only increase precision. For the Gaussian family the expectation is exact and (S23) coincides with (S20); for the other families it is the expected-information (Fisher-scoring) form used in §S7. $\square$

S4.2 Per-family unit information

Because η_{ij} is a single index in f_i with $\nabla_{f_i} \eta_{ij} = \lambda_j$ (Proposition 7), the information that item j carries about the factors factorizes through the scalar index:

$$\mathcal{I}_j(f_i) = \mathcal{J}_j(\eta_{ij}) \lambda_j \lambda_j^\top, \quad \mathcal{J}_j(\eta) = -\mathbb{E}_{X|\eta} \left[\frac{\partial^2}{\partial \eta^2} \log F_j(X | \eta) \right], \quad (\text{S24})$$

so the per-factor information is $\mathcal{I}_{jk}(f_i) = \lambda_{jk}^2 \mathcal{J}_j(\eta_{ij})$ — Eq. (9) of the main text — with $\mathcal{J}_j$ the one-dimensional Fisher information of the item's own likelihood family. We derive $\mathcal{J}_j$ for each family.

Proposition 10 (Family unit informations). *For the five families of (S6)–(S10),*

$$\text{Gaussian: } \mathcal{J}_j(\eta) = \frac{1}{\sigma_j^2}, \quad (\text{S25})$$

$$\text{Bernoulli (logit): } \mathcal{J}_j(\eta) = p(1-p), \quad p = \text{logit}^{-1}(\eta), \quad (\text{S26})$$

$$\text{Negative-binomial (log link): } \mathcal{J}_j(\eta) = \frac{\mu r_j}{\mu + r_j}, \quad \mu = e^\eta, \quad (\text{S27})$$

$$\text{Graded-response ordinal: } \mathcal{J}_j(\eta) = \sum_{c=1}^{C_j} \frac{(\partial_\eta P_{jc}(\eta))^2}{P_{jc}(\eta)} = \sum_{c=1}^{C_j-1} \frac{(P_{jc}^*(1 - P_{jc}^*) - P_{j,c+1}^*(1 - P_{j,c+1}^*))^2}{P_{jc}(\eta)}, \quad (\text{S28})$$

where $P_{jc}(\eta) = P_{jc}^* - P_{j,c+1}^*$ is the probability of category c and $P_{jc}^* = \text{logit}^{-1}(\eta - \tau_{jc})$ is the cumulative probability of exceeding threshold τ_{jc} (with $P_{j0}^* = 1$, $P_{j,C_j}^* = 0$).

Proof. Gaussian. $\log F_j = -\frac{1}{2} \log(2\pi\sigma_j^2) - \frac{(X-\eta)^2}{2\sigma_j^2}$, so $\partial_\eta^2 \log F_j = -1/\sigma_j^2$, a constant; its negative expectation is $1/\sigma_j^2$, independent of η .

Bernoulli. With $p = \text{logit}^{-1}(\eta)$, $\log F_j = X \log p + (1-X) \log(1-p)$ and $\partial_\eta p = p(1-p)$. The score is $\partial_\eta \log F_j = X - p$, and $\mathcal{J}_j = \text{Var}(X - p | \eta) = \text{Var}(X) = p(1-p)$.

Negative-binomial. For $X \sim \text{NegBinom}$ with mean $\mu = e^\eta$ and dispersion r_j , $\text{Var}(X | \eta) = \mu + \mu^2/r_j$. With the log link $\partial_\eta \mu = \mu$, the generalized-linear-model information is $\mathcal{J}_j = (\partial_\eta \mu)^2 / \text{Var}(X) = \mu^2 / (\mu + \mu^2/r_j) = \mu r_j / (\mu + r_j)$.

Graded-response ordinal. For a categorical likelihood $\log F_j = \sum_c \mathbf{1}\{X = c\} \log P_{jc}(\eta)$, the score is $\partial_\eta \log F_j = \sum_c \mathbf{1}\{X = c\} \partial_\eta P_{jc} / P_{jc}$ and the expected squared score is $\mathcal{J}_j = \sum_c (\partial_\eta P_{jc})^2 / P_{jc}$, the standard graded-response information [1]. Writing category probabilities as differences of the cumulative logits P_{jc}^* and using $\partial_\eta P_{jc}^* = P_{jc}^* (1 - P_{jc}^*)$ gives the closed form on the right of (S28). $\square$

S4.3 Why loading is not information

Equation (S24) is why localization and reflection part company. The loading λ_{jk} enters $\mathcal{I}_{jk}$ squared, but it multiplies the unit information $\mathcal{J}_j(\eta_{ij})$, which for every non-Gaussian family depends on where the patient sits. The Bernoulli case (S26) is the sharpest: a binary flag endorsed by a fraction p of patients contributes $\mathcal{J}_j = p(1-p)$, which is 0.09 at $p = 0.1$ and collapses to 0 as $p \rightarrow 0$, *regardless of* λ . A strongly-loading, rarely-endorsed symptom therefore localizes almost no one. Only for the Gaussian family (S25), where $\mathcal{J}_j = 1/\sigma_j^2$ is constant, does the ranking by information coincide with the ranking by (scaled) loading.

Corollary 2 (Reliability and the target-driven item count). *On a single axis k , accumulating (S23) gives a scalar precision $\pi_k = 1 + \sum_j \mathcal{I}_{jk}$ (prior precision 1 for the standardized factor) and a reliability*

$$\rho_k = 1 - \frac{1}{\pi_k} = 1 - \frac{1}{1 + \sum_j \lambda_{jk}^2 \mathcal{J}_j}. \quad (\text{S29})$$

To reach a posterior standard deviation $\pi_k^{-1/2} \leq \varsigma$ requires accumulated information $\sum_j \mathcal{I}_{jk} \geq \varsigma^{-2} - 1$; at $\varsigma = 0.5$ this is $\sum_j \mathcal{I}_{jk} \geq 3$. For a standardized continuous indicator of common loading λ the residual variance is $1 - \lambda^2$, so its Gaussian unit information is $\mathcal{J} = 1/(1 - \lambda^2)$ and it contributes $\lambda^2/(1 - \lambda^2)$ per item; the required count is therefore

$$n \geq \frac{\varsigma^{-2} - 1}{\lambda^2/(1 - \lambda^2)} = 3 \frac{1 - \lambda^2}{\lambda^2} \quad (\varsigma = 0.5), \quad (\text{S30})$$

i.e. 3.1 items at $\lambda = 0.7$, 9.0 at $\lambda = 0.5$ and 30.3 at $\lambda = 0.3$ — consistent with the “roughly 4, 9 and 31 items” the full mixed-bank simulation reports in §3.2.

Proof. Immediate from $\pi_k = 1 / \text{Var}(f_{ik} | X)$ and (S29): the item count solves $\lambda^2 \mathcal{J} \cdot n = \varsigma^{-2} - 1$ with the standardized-indicator unit information $\mathcal{J} = 1/(1 - \lambda^2)$, giving (S30). $\square$

S5 Adaptive sampling and value of information

S5.1 The fixed order as an upper bound on adaptive efficiency

Ordering the candidate items for an axis by their information (S24) and accumulating (S23) gives the reliability-versus-items curve (S29). Evaluated at the population mean $f_i = \mathbf{0}$ this is the efficiency of a fixed, non-adaptive most-informative-first battery. A computerized adaptive test (CAT) instead re-evaluates $\mathcal{J}_j(\eta_{ij})$ at each patient’s provisional estimate $\hat{\theta}$ and selects the currently most informative item.

Proposition 11 (Population-mean order is an upper bound). *Fix an axis and let $\bar{\mathcal{I}}_j = \mathcal{J}_j(\bar{\eta}_j)$ be the unit information evaluated at the population-mean linear predictor $\bar{\eta}_j = \alpha_j$. The fixed schedule that administers items in decreasing $\lambda_j^2 \bar{\mathcal{I}}_j$ maximizes accumulated information at every step among schedules that do not use the patient’s responses. Any adaptive schedule attains, in expectation over patients, no more accumulated information per item at the population mean than this fixed order.*

Proof. At the population mean the per-item increments $\lambda_j^2 \bar{\mathcal{I}}_j$ are fixed constants; by the rearrangement inequality, administering them in decreasing order maximizes the partial sums $\sum_{j \leq n}$ at every n , so the fixed most-informative-first order is optimal among response-independent schedules. An adaptive schedule can exceed a fixed one only by exploiting the dependence of $\mathcal{J}_j(\eta_{ij})$ on the unknown f_i ; at the population mean, and for families whose $\mathcal{J}_j$ does not depend on η (the Gaussian family, (S25)), that dependence is absent and the two schedules coincide. $\square$

This is why the CAT simulation of §3.3 finds the bound *tight*: on every axis whose leading items are Gaussian the adaptive and fixed curves coincide to within 2×10^{-4} in reliability, because $\mathcal{J}_j$ is constant and there is nothing to adapt to; only Bernoulli-topped axes (whose $\mathcal{J}_j = p(1-p)$ varies with θ) show a small adaptive signature, at most 0.012 in reliability. The bank-limited axes are a separate phenomenon: mania/activation and substance plateau at reliability 0.41 and 0.43 because $\sum_j \lambda_{jk}^2 \mathcal{J}_j$ saturates over a 2- and 4-item bank respectively — a ceiling of (S29) set by instrument content, not by any schedule.

S5.2 Value of information: the matrix-determinant lemma

Across axes the same information becomes a design criterion through the entropy of the posterior coordinate.

Proposition 12 (Closed-form entropy reduction). *The differential entropy of the Gaussian posterior is $H(f_i) = \frac{1}{2} \log \det(2\pi e S_i)$. Adding a single not-yet-measured indicator j^* with loading λ_{j^*} and unit information $\mathcal{J}_{j^*} = \mathcal{J}_{j^*}(\hat{\eta}_{ij^*})$ reduces the entropy by*

$$H(f_i) - H(f_i | j^*) = \frac{1}{2} \log \left(1 + \mathcal{J}_{j^*} \lambda_{j^*}^\top S_i \lambda_{j^*} \right), \quad (\text{S31})$$

a closed form that depends on what has already been measured through the current posterior covariance S_i . Equation (S31) is Eq. (10) of the main text.

Proof. By Proposition 9 the updated precision is $S_i^{-1} |_{+j^*} = S_i^{-1} + \mathcal{J}_{j^*} \lambda_{j^*} \lambda_{j^*}^\top$, a rank-one update. The entropy change is $H(f_i) - H(f_i | j^*) = \frac{1}{2} \log \det(S_i^{-1} |_{+j^*}) - \frac{1}{2} \log \det(S_i^{-1}) = \frac{1}{2} \log \frac{\det(S_i^{-1} + \mathcal{J}_{j^*} \lambda_{j^*} \lambda_{j^*}^\top)}{\det S_i^{-1}}$. The matrix-determinant lemma $\det(A + uv^\top) = \det(A)(1 + v^\top A^{-1}u)$ with $A = S_i^{-1}$, $u = \mathcal{J}_{j^*} \lambda_{j^*}$, $v = \lambda_{j^*}$ gives the ratio $1 + \mathcal{J}_{j^*} \lambda_{j^*}^\top S_i \lambda_{j^*}$, and the logarithm halves to (S31). $\square$

Two properties make (S31) the right design criterion. It is *context-dependent*: the value of a candidate item scales with $\lambda_{j^*}^\top S_i \lambda_{j^*}$, the current posterior variance along the item’s loading direction, so an item is worth most on an axis that is still uncertain and its value falls as that axis fills in. And it is *diminishing*: because S_i shrinks with each added item, the marginal entropy reduction of the next item decreases, so a greedy rule spreads measurement across axes rather than piling onto one. Cycling the greedy selection across the eight axes assembles the shared 27-item battery at mean reliability 0.70 of §3.4; applied within a cohort, the same S_i -weighted criterion localizes where an axis is under-measured (e.g. the schizophrenia immunometabolic and sleep deficits, 20.7 vs 34.9 and 3.9 vs 8.4 items).

Corollary 3 (Greedy value is a lower bound on the jointly-optimal design). *The entropy reduction $f \mapsto \log \det(\Phi^{-1} + \sum_{j \in S} \mathcal{I}_j)$ as a function of the selected set S is monotone and submodular, so the greedy battery attains at least a $(1 - 1/e)$ fraction of the entropy reduction of the jointly-optimal same-size battery.*

Proof. Monotonicity holds because each $\mathcal{I}_j \succeq 0$ (Proposition 9) can only increase the log-determinant. Submodularity of $S \mapsto \log \det(A + \sum_{j \in S} \mathcal{I}_j)$ for positive semidefinite $\mathcal{I}_j$ is the standard concavity-of-log-det result: the marginal gain of adding j^* , given by (S31), is non-increasing as the base set grows because S_i (the inverse of the accumulated precision) is decreasing

in the Loewner order. The $(1 - 1/e)$ guarantee is the Nemhauser–Wolsey–Fisher bound for greedy maximization of a monotone submodular function. This bounds the price of greediness noted as a limitation in the main text. $\square$

S6 Archetypes as a low-dimensional summary and its reconstruction fidelity

S6.1 The archetypal-analysis program

The cloud of posterior coordinates $\{x_i\}_{i=1}^N$, $x_i = \hat{f}_i \in \mathbb{R}^{K+1}$, is summarized by archetypal analysis [2]: each patient is written as a convex blend of A extreme phenotypes that are themselves convex combinations of patients,

$$\min_{W,B} \sum_{i=1}^N \left\| x_i - \sum_{a=1}^A w_{ia} z_a \right\|^2 \quad \text{s.t.} \quad z_a = \sum_{j=1}^N b_{ja} x_j, \quad w_{ia}, b_{ja} \geq 0, \quad \sum_a w_{ia} = \sum_j b_{ja} = 1, \quad (\text{S32})$$

i.e. $X \approx WZ$ with archetypes $Z = BX$ and both W (patient $\times$ archetype) and B (patient $\times$ archetype) row-stochastic. This is Eq. (11) of the main text. The constraints place each archetype z_a inside the convex hull of the data and each patient’s reconstruction $\hat{x}_i = \sum_a w_{ia} z_a$ inside the convex hull of the archetypes; the simplex weight $w_i \in \Delta^{A-1}$ is the patient’s soft membership. The number of corners is fixed by stability: $A = 5$ is the largest with cross-seed Tucker congruence ≥ 0.8 , with a cliff at $A = 6$, and w_i is propagated across posterior draws.

S6.2 Reconstruction fidelity as a variance decomposition

Because (S32) is a rank- A convex approximation it discards information, and unlike an unconstrained projection its residual is not orthogonal to the fitted part. We quantify the loss by a coefficient of determination.

Definition 1 (Reconstruction R^2). With reconstruction $\hat{x}_i = \sum_a w_{ia} z_a$, the per-axis and pooled reconstruction R^2 are

$$R_k^2 = 1 - \frac{\sum_i (x_{ik} - \hat{x}_{ik})^2}{\sum_i (x_{ik} - \bar{x}_k)^2}, \quad R_{\text{pool}}^2 = 1 - \frac{\sum_i \|x_i - \hat{x}_i\|^2}{\sum_i \|x_i - \bar{x}\|^2} = \frac{\sum_k \text{SS}_k^{\text{tot}} R_k^2}{\sum_k \text{SS}_k^{\text{tot}}}, \quad (\text{S33})$$

the pooled value being the variance-weighted mean of the per-axis values with weights $\text{SS}_k^{\text{tot}} = \sum_i (x_{ik} - \bar{x}_k)^2$.

Evaluated on all $N = 9,013$ patients the five-corner blend attains $R_{\text{pool}}^2 = 0.590$: the convex summary retains 59% of the coordinate variance. It is tempting to read this against the unconstrained linear optimum at the same dimension — a five-component principal-component projection attains $R^2 = 0.796$, a dimension-matched four-component projection 0.680 — and conclude that convex interpretability costs 20.6 percentage points of fidelity. That reading is wrong, and Proposition 13 below explains why: on this map the coordinate variance is nearly isotropic, so reconstruction R^2 is an uninformative test of any five-dimensional summary. A *random* five-dimensional subspace retains an expected $k/d = 5/8 = 0.625$ of the variance regardless of the covariance spectrum — more than the archetype simplex — and a 500-seed simulation confirms it (0.624 ± 0.042 , placing the archetype value at the 21st percentile of random projections). The simplex does not merely trail PCA; it trails a coin-flip subspace, because variance capture is not what it optimizes. What it does provide, and what neither PCA nor a random subspace can, is convex non-negative sum-to-one membership in A *named* corners: a principal component is a signed axis-mixing contrast that no patient can equal, whereas each archetype z_a is a near-real extreme profile a patient can be a stated fraction of. Reconstruction

R^2 is therefore reported here as an honesty check on how much of the continuum the corners span — not as a competitive metric — and the five-centroid k -means hard partition ($R^2 = 0.325$), the discrete alternative appropriate only if the cloud had clusters, is included to show that the least defensible summary is the categorical one. Per axis, R_k^2 is high where the corners separate (immunometabolic 0.89, overall severity 0.86, sleep 0.76, developmental risk 0.75) and near-flat on the two bank-limited axes (cognition 0.06, substance ≈ 0), the same axes flagged by (S29) — so the per-axis profile is a diagnostic of where the instrument is weak, not a scorecard.

Proposition 13 (A random k -subspace captures k/d of the variance). *Let $x_1, \dots, x_N \in \mathbb{R}^d$ have mean zero and total variance $T = \frac{1}{N} \sum_i \|x_i\|^2$. Let P be the orthogonal projector onto a k -dimensional subspace drawn uniformly (Haar) from the Grassmannian $\text{Gr}(k, d)$. Then the expected variance captured by projecting the data onto that subspace is*

$$\mathbb{E}\left[\frac{1}{N} \sum_i \|Px_i\|^2\right] = \frac{k}{d} T, \quad \text{i.e.} \quad \mathbb{E}[R_{\text{rand-}k}^2] = \frac{k}{d}, \quad (\text{S34})$$

independent of the eigenvalue spectrum of the data covariance.

Proof. Write the covariance $\Sigma = \frac{1}{N} \sum_i x_i x_i^\top$ with eigenvalues $\sigma_1^2 \geq \dots \geq \sigma_d^2$ summing to T . The captured variance is $\frac{1}{N} \sum_i x_i^\top P x_i = \text{tr}(P\Sigma)$. For a Haar-random rank- k projector, $\mathbb{E}[P] = (k/d)I_d$ by rotational invariance (the only symmetric matrix commuting with all rotations is a scalar multiple of the identity, and $\text{tr} \mathbb{E}[P] = k$ fixes the scalar). Hence $\mathbb{E}[\text{tr}(P\Sigma)] = \text{tr}((k/d)\Sigma) = (k/d)T$, and dividing by T gives (S34). The result uses only $\mathbb{E}[P] = (k/d)I$; the spectrum $\{\sigma_j^2\}$ cancels, which is why an anisotropic (e.g. single-dominant-component) cloud and a near-isotropic one both give k/d in expectation, while their PCA-optimal R^2 differ sharply. $\square$

For this map $k/d = 5/8 = 0.625$; the simulated mean 0.624 matches to Monte-Carlo error, and the gap between that and the PCA-5 value 0.796 is a direct measure of how far the coordinate cloud departs from isotropy — here, modestly (the leading component carries 0.269 of the variance, against 0.125 for an exactly isotropic cloud).

Proposition 14 (Fidelity does not degrade for ambiguous patients). *Let $e_i = \|x_i - \hat{x}_i\|$ be a patient’s reconstruction error and $H_i = -\sum_a w_{ia} \log w_{ia}$ the entropy of their archetype weights. Then $\text{Corr}(e_i, H_i) < 0$: interior “fog” patients reconstruct better, not worse.*

Argument. By construction each $\hat{x}_i$ lies in the convex hull of the archetypes. A high-entropy patient has weights spread across corners and therefore sits near the centroid of the hull, where the convex approximation is tight; a low-entropy (near-corner) patient sits near a single archetype z_a , but the true x_i near a corner frequently lies *outside* the hull spanned by the finite archetype set (the archetypes are interior convex combinations of data, not the extreme points themselves), so the clipped reconstruction incurs a larger residual. Empirically $\text{Corr}(e_i, H_i) = -0.13$ ($p < 10^{-30}$), consistent with this geometry. The negative sign is the useful conclusion: the summary is not purchased by sacrificing the ambiguous patients it is meant to describe. $\square$

S7 Calibration protocol

The instrument reports posterior intervals; §2.7 tests whether those intervals are honest. Because the truth is unknown for real patients, coverage is tested by simulation under the model’s own generative law with real missingness.

S7.1 Synthetic-truth generative design

We draw $M = 10,000$ synthetic patients (seed 20260704) with known latent coordinate $\theta_m \sim \mathcal{N}(\mathbf{0}, \Phi)$ under the frozen model’s factor prior. For each we generate item responses from

the fitted families (S6)–(S10) at $\eta_{mj} = \alpha_j + \lambda_j^\top \theta_m$, and impose a real missingness pattern by bootstrapping the observed/missing masks of the 9,013 real patients — only the binary observed-or-missing indicator is used, never any clinical value. This preserves the empirical joint distribution of *which* axes are co-measured, which is what stresses the projection.

S7.2 Fisher-scoring EAP and its exactness on Gaussian patterns

Each synthetic patient is projected by the same machinery as a real one. For non-Gaussian patterns the posterior mode and curvature are found by Fisher scoring: starting from $\hat{\theta}^{(0)} = \mathbf{0}$, iterate

$$\hat{\theta}^{(t+1)} = \hat{\theta}^{(t)} + \left(\Phi^{-1} + \sum_{j \in O_m} \mathcal{I}_j(\hat{\theta}^{(t)}) \right)^{-1} \left(-\Phi^{-1} \hat{\theta}^{(t)} + \sum_{j \in O_m} u_{mj}(\hat{\theta}^{(t)}) \right), \quad (\text{S35})$$

where $u_{mj} = \lambda_j \partial_\eta \log F_j(X_{mj} | \eta_{mj})$ is the item score and $\mathcal{I}_j$ its expected information (S24); the posterior covariance is the inverse information $S_m = (\Phi^{-1} + \sum_j \mathcal{I}_j(\hat{\theta}))^{-1}$ at convergence, the Laplace approximation to the posterior.

Proposition 15 (Exactness on Gaussian patterns). *If all observed items of patient m are Gaussian (S6), the Fisher-scoring iteration (S35) converges in one step from any start to the closed-form EAP (S19), and S_m equals the closed-form posterior covariance exactly.*

Proof. For a Gaussian item $\mathcal{I}_j = \lambda_j \lambda_j^\top / \sigma_j^2$ is independent of θ and the score is affine, $u_{mj}(\theta) = \lambda_j (X_{mj} - \alpha_j - \lambda_j^\top \theta) / \sigma_j^2$. The bracketed Newton system (S35) is then exactly the normal equations of the Gaussian posterior, whose solution is (S20)–(S19); a Newton step on a quadratic objective reaches the optimum in one iteration regardless of the starting point. $\square$

We verified this numerically: on all-Gaussian patterns the Fisher-scoring coordinate matches the closed form (S19) to 3×10^{-15} , so the calibration study exercises the *same* estimator the instrument uses, extended to the non-Gaussian families.

S7.3 Coverage statistics and the prior-in-prior-out check

For nominal levels $q \in \{0.50, 0.80, 0.90, 0.95\}$ with standard-normal quantile z_q , we report three statistics: the per-axis marginal coverage

$$\widehat{\text{cov}}_k(q) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\left\{ |\theta_{mk} - \hat{x}_{mk}| \leq z_{(1+q)/2} [S_m]_{kk}^{1/2} \right\}, \quad (\text{S36})$$

the joint coverage of the 8-dimensional credible ellipsoid via the Mahalanobis distance against a χ_{K+1}^2 quantile, $\widehat{\text{cov}}_{\text{joint}}(q) = \frac{1}{M} \sum_m \mathbf{1}\{(\theta_m - \hat{x}_m)^\top S_m^{-1}(\theta_m - \hat{x}_m) \leq \chi_{K+1,q}^2\}$, and the 95% coverage stratified by the number of observed indicators on the axis. The Monte-Carlo standard error of a coverage estimate at level q is $\sqrt{q(1-q)/M}$; at $q = 0.95$, $M = 10^4$ this is 0.0022.

Proposition 16 (Prior in, prior out). *On an axis k with no observed indicator, the projection returns the prior on that coordinate — $\hat{x}_{mk} = 0$ and $[S_m]_{kk}$ equal to the prior variance — so the coverage (S36) equals the nominal level exactly, up to Monte-Carlo error.*

Proof. With no observed item loading on axis k , the corresponding rows of Λ_{O_m} are zero, so the data term $\sum_j \mathcal{I}_j$ in (S23) has no entry on coordinate k and the posterior marginal of θ_{mk} equals its prior $\mathcal{N}(0, \Phi_{kk})$. Since the synthetic truth is drawn from that same prior (S1), the interval $0 \pm z_{(1+q)/2} \Phi_{kk}^{1/2}$ covers θ_{mk} with probability exactly q . $\square$

This makes the zero-item stratum a built-in correctness check rather than a free parameter: the instrument *must* recover nominal coverage there, and it does. Across all strata the empirical coverage tracks nominal — 0.501, 0.801, 0.900, 0.949 against 0.50/0.80/0.90/0.95, joint ellipsoid 0.947 at 95% — and is flat from the prior-dominated stratum to the richly-observed one (0.947–0.952 across item-count bins), the result reported in §3.5.

What the study does and does not establish. Coverage here demonstrates that the projection machinery is *internally honest*: its 95% intervals cover the truth 95% of the time when the data are generated by the fitted model under real missingness. It does not establish that the fitted model is the true data-generating process — no coverage study can, since it conditions on the generative law. On the discrete-informed axes the Laplace posterior standard deviation runs slightly below the residual error (a light-tailed approximation), so interval *width* is marginally optimistic even though the 95%-level coverage, fixed by the interval quantiles, is exact. The next subsection addresses the conditioning limitation directly.

S7.4 Robustness of coverage under misspecification

The coverage study of §S7.3 draws data from the fitted model, so it cannot distinguish a well-calibrated estimator from one that is calibrated only because it scores data it also generated. We therefore stress the projection with data drawn from processes it does *not* assume, and ask how far coverage drifts. Four one-parameter families of violations are used, each targeting a specific assumption of §S1:

1. **Correlated residuals** (violates local independence, (S4)): the residual of every item receives a shared nuisance component, $\varepsilon_j \leftarrow (\varepsilon_j + \rho g_*) / \sqrt{1 + \rho^2}$ with common $g_* \sim \mathcal{N}(0, 1)$ and $\rho \in \{0, 0.15, 0.30, 0.45\}$.
2. **Omitted factor** (violates the rank- K assumption): a ninth latent $g_9 \sim \mathcal{N}(0, 1)$ loads a random 40% of items with strength $s \in \{0, 0.25, 0.5, 0.75\}$, absent from the scoring model.
3. **Loading mismatch** (parameter error / transfer to a new sample): the scoring model uses $\lambda_{jk}(1 + \varepsilon\xi_{jk})$, $\xi_{jk} \sim \mathcal{N}(0, 1)$, $\varepsilon \in \{0, 0.10, 0.20, 0.30\}$, while data use the true λ_{jk} .
4. **Heavy tails** (violates Gaussian residuals): continuous residuals are Student- t with $\text{df} \in \{\infty, 8, 5, 3\}$, rescaled to unit variance.

Each cell draws $N = 4,000$ synthetic patients keeping 50% of items (three seeds), scores them with the *unperturbed* Gaussian EAP, and records empirical coverage of the nominal 95% per-axis interval. The results (Fig. 9B; §3.8) show graceful, monotone degradation and no collapse:

Violation	none	low	med	high
Correlated residuals (ρ)	0.951	0.943	0.918	0.889
Omitted ninth factor (s)	0.951	0.948	0.939	0.939
Loading mismatch (ε)	0.949	0.950	0.927	0.934
Heavy-tailed residuals (df)	0.949	0.950	0.950	0.955

Three features are worth noting. First, the unperturbed baseline recovers nominal (0.949–0.951) in every family, confirming the estimator is unbiased when its assumptions hold. Second, heavy tails are absorbed essentially without loss — indeed slightly conservatively at the 50% level — because the EAP posterior variance depends on the loadings and the residual *variance*, not on higher residual moments. Third, the steepest degradation is under correlated residuals, and this is expected: local dependence inflates the effective information a naive independent-likelihood score believes it has, shrinking intervals below their honest width. Even so, an extreme shared-nuisance loading ($\rho = 0.45$, a residual correlation the model-selection diagnostics of the main text would flag) costs only six points of coverage. The projection’s intervals are thus honest under the model and degrade predictably, not catastrophically, away from it; local dependence is the assumption whose violation matters most, and the one most worth checking on real data before trusting interval *widths*.

Summary of cross-references to the main text

Each numbered result here underlies a specific claim in the main manuscript: (S11) and Proposition 1 justify the no-imputation invariant (Eq. (3)); Lemma 1 and Proposition 4 are the projection (Eq. (6), Eq. (7)); Propositions 7–8 are the loading-as-slope (Eq. (5)); Propositions 9–10 and Corollary 2 are the information accounting (Eq. (8), Eq. (9)); Propositions 11–12 and Corollary 3 are adaptive sampling and value of information (Eq. (10)); Definition 1, Proposition 13 (the random-subspace k/d bound that reframes reconstruction R^2) and Proposition 14 are the archetype reconstruction; and Propositions 15–16 with the misspecification stress test of §S7.4 are the calibration protocol and its robustness (§2.7, §3.8).

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